

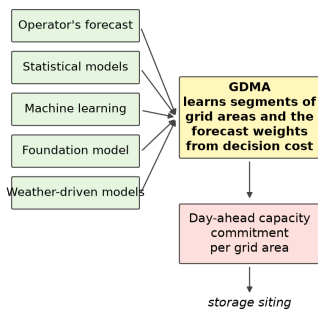
Graphical Abstract

From Forecasts to Batteries: Grouped Decision-Focused Model Averaging for Capacity Commitment and Storage Siting on Five Continents

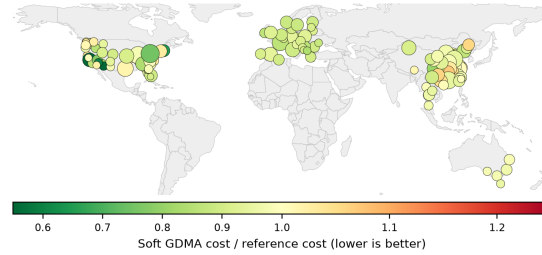
Quang-Vinh Dang

Grouped decision-focused model averaging for capacity commitment and storage siting

1 Many forecasts, one decision



2 125 grid areas, 5 continents, public data 2018-2026



3 Results (out of sample)

-12% commitment cost vs reference forecaster, worldwide

-4 pp vs each area's best single forecast

34 GW of batteries the reference rule needs to match soft GDMA

6% of areas worse off than their own best candidate

Highlights

From Forecasts to Batteries: Grouped Decision-Focused Model Averaging for Capacity Commitment and Storage Siting on Five Continents

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- Grouped model averaging learns forecast weights from commitment cost, not accuracy.
- Soft GDMA cuts day-ahead commitment cost by 12% on 125 grid areas, five continents.
- Oracle inequality and segment recovery proved for the fixed- G estimator.
- 6% of areas are worse off; no systematic income gradient is found.
- Better commitment rules lower the value of storage and change where to site it.

From Forecasts to Batteries: Grouped Decision-Focused Model Averaging for Capacity Commitment and Storage Siting on Five Continents

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Abstract

Grid operators commit capacity a day ahead from many demand forecasts, and planners must decide where to install batteries. We propose *grouped decision-focused model averaging* (GDMA): it learns segments of grid areas and segment-specific forecast weights by minimising each area’s realised commitment cost, not forecast error. We prove a finite-sample oracle inequality and segment recovery for the fixed- G estimator. On 125 grid areas on five continents, using public data for 2018–2026 and only information available at gate closure, soft GDMA cuts commitment cost by 12% against a reference forecaster (13–17% against the operators’ own forecasts in the United States) and by 4 percentage points against each area’s best single forecast. A decision-focused neural gate costs 7 percentage points more, and grouping adds little over shrinkage. Soft GDMA leaves 6% of areas worse off and shows no systematic income gradient. Weather forecasts halve the gap to the operators during Winter Storm Elliott but do not close it. In a stylised storage model, the reference rule needs 34 GW of batteries to match soft GDMA without any. Lower cost is better throughout.

Keywords: model averaging, decision-focused learning, latent groups, foundation models, fairness, capacity commitment, battery storage siting, energy management

1. Introduction

Electricity systems consist of many local decision units that commit resources before uncertainty resolves. Every day, each balancing authority and market operator must commit enough capacity for the next day’s hourly demand: too little forces expensive real-time purchases, reserve activation or load shedding, too much pays for idle capacity. Shortfalls cost less for a well-interconnected operator, which can import from neighbours, than for an isolated one. In Winter Storm Uri in February 2021 the weakly interconnected Texas grid shed load for days, while neighbours with more import capability shed much less (Busby et al., 2021; FERC, NERC and Regional Entity Staff, 2021). Meanwhile, governments are funding large battery programmes, and planners must decide where the batteries go.

Forecasts are plentiful: official day-ahead forecasts, persistence and calendar rules, statistical and machine-learning models, and foundation models that forecast any series

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zero-shot (Ansari et al., 2024). None dominates everywhere, so combining them is attractive. Model averaging, from Bates and Granger (1969) to Hansen (2007), Hansen and Racine (2012), Zhang et al. (2013) and Zhang and Liu (2023), offers two options for many series. Weights can be estimated series by series, which respects heterogeneity but is noisy when histories are short—the root of the “forecast-combination puzzle” that simple averages often beat estimated weights (Smith and Wallis, 2009; Claeskens et al., 2016)—or one set can be estimated for all series, which is precise but imposes one rule on systems facing different conditions. A third option, a network mapping features to weights (Montero-Manso et al., 2020; Gong et al., 2026), costs thousands of parameters and weights that are hard to audit. All three typically judge a combination by accuracy, whereas the planner is judged by decision cost; learning directly for the decision can be much better than forecasting accurately and then optimising (Ban and Rudin, 2019; Bertsimas and Kallus, 2020; Elmachoub and Grigas, 2022).

This paper proposes *grouped decision-focused model averaging* (GDMA) and uses it to answer three management questions: how should grid operators on five continents combine their forecasts into day-ahead capacity commitments, who gains and who loses from a common planning rule, and where should new batteries be installed? GDMA assumes that decision units fall into a small number of latent segments whose members are best served by the same combination of candidate rules, and estimates segments and segment-specific weights jointly by minimising the realised system-wide decision cost, in money, with each unit’s own cost asymmetry. A soft version shrinks each unit’s weights towards its segment’s, and the number of segments and the shrinkage intensity are chosen by a time-ordered hold-out. Pooled weights, unit-specific weights and shrinkage towards pooled weights are special cases, and each commitment is a convex combination of commitments the operator already understands. A second stage allocates a storage budget across grid areas using the residual shortfall risk of the learned rule.

Figure 1 summarises the approach and Figure 2 the timing of the information.

We make five contributions.

1. **Method.** We formulate joint learning of latent segments and combination weights under an asymmetric, unit-specific decision loss, give a k -means-type algorithm with regret-space seeding in which every step is a small convex program, and link it to a storage-siting problem solved exactly by a greedy algorithm (Section 3). To our knowledge, no model-averaging, decision-focused or latent-group panel work (Bonhomme and Manresa, 2015; Su et al., 2016) proposes this estimator.
2. **Learning theory.** We prove a finite-sample oracle inequality that separates the cost of learning memberships ($\log G/T$ per unit) from the cost of learning weights ($GM \log(NT)/(NT)$); GDMA is asymptotically optimal, attains under a segment structure the risk of the infeasible benchmark in which every unit knows its own optimal weights, recovers the segments with probability approaching one, and then coincides exactly with an estimator told the segments (Section 4).
3. **Evidence on five continents.** With information available at the day-ahead gate closure, on 46 U.S. balancing authorities with their official forecasts (Section 6) and on a global panel of 125 grid areas in North America, Europe, Oceania, Asia and Africa (Section 7), soft GDMA lowers the stylised out-of-sample commitment cost by 13–17% relative to the official forecasts in the United States and by 12% worldwide relative to a reference forecaster, beats accuracy-based combination by 2–3 points in the United States (at $W = 28$) and 8–10 points worldwide, and beats a decision-focused neural mixture-of-experts gate by seven percentage points

A. From public data to storage siting

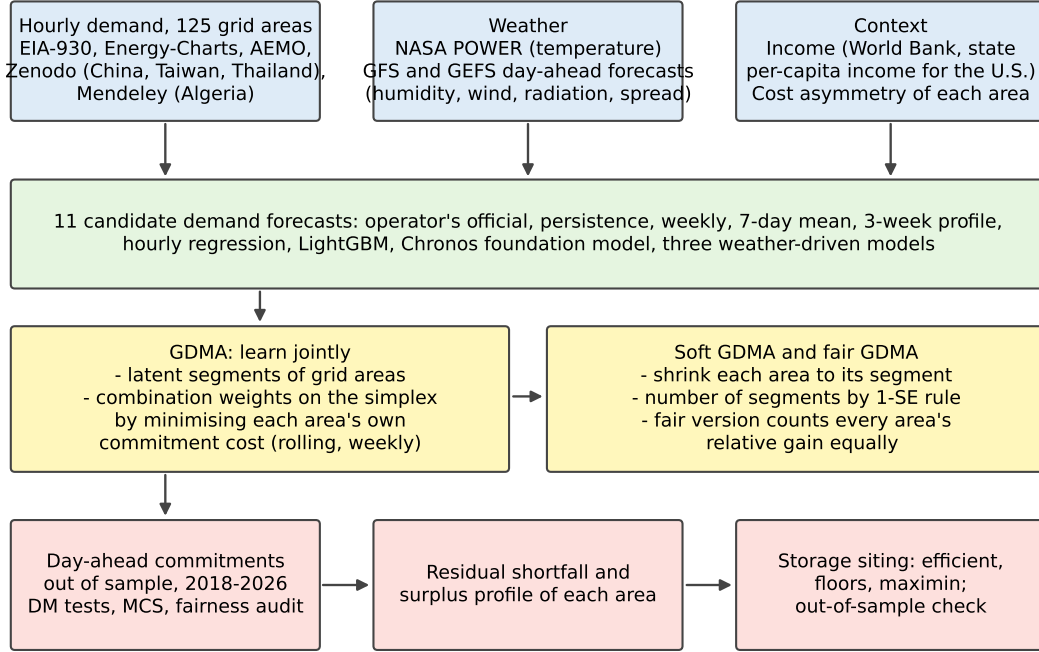


Figure 1: The approach from public data to storage siting.

worldwide. Decision-focused shrinkage and two-step clustering come close and a model confidence set cannot separate the grouped variants, so the evidence favours decision-focused combination with cross-sectional structure, not grouping alone. Ensemble weather information roughly halves, but does not close, the gap to the operators' forecasts during Winter Storm Elliott, the most abrupt event in the sample.

4. **Fairness.** We audit how gains are distributed and propose a fair variant that counts every area's relative improvement equally. Soft GDMA makes 6% of areas worse off than their own best candidate, against 24% for pooled weights and 42% for the neural gate, shows no detectable gradient with national or state income, and its fair variant halves the share of areas made worse off at no cost in total (Section 7.4).
5. **Storage siting.** In a stylised storage model, better commitment rules reduce what storage can add: the reference rule needs 34 GW of optimally sited batteries to reach the cost that soft GDMA achieves without any. Siting on the wrong rule's error profile loses only 2–4% of a programme's value, and maximin-fair siting helps the worst-served continent at a price of fairness of 7–9% (Section 8).

The paper fits the theme of this special issue in both directions: a statistical learning method with guarantees, motivated by a management problem, and a continental-scale planning question answered with open data. Replication code is in a public repository (see the data availability statement).

2. Related literature and the research gap

Model averaging and forecast combination. Since [Bates and Granger \(1969\)](#), combining forecasts has been one of the most reliable ways to improve accuracy; [Timmermann](#)

B. Information set for the commitment of UTC day d

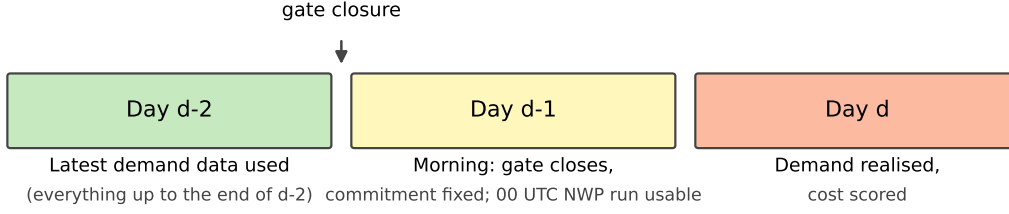


Figure 2: Information set of the day-ahead commitment. Data up to the end of day $d - 2$ and the 00 UTC weather forecast of day $d - 1$ (only where it is available before gate closure) enter the commitment for day d .

(2006) and Wang et al. (2023) review five decades. Frequentist model averaging chooses weights by minimising estimated risk—a Mallows criterion (Hansen, 2007), leave-one-out or K -fold cross-validation (Hansen and Racine, 2012; Zhang and Liu, 2023), or forward validation for time series (Zhang and Zhang, 2023)—and establishes asymptotic optimality: the chosen weights attain the lowest attainable risk. It now covers dependent data (Zhang et al., 2013), quantile losses (Lu and Su, 2015), general losses (Yu et al., 2024), time-varying weights (Sun et al., 2021), covariate-dependent weights (Gong et al., 2026), covariate shift (Zhang et al., 2026) and prediction intervals (Qu et al., 2025). The forecast-combination puzzle, that equal weights frequently beat estimated weights, is attributed mainly to estimation error (Smith and Wallis, 2009; Claeskens et al., 2016); remedies shrink weights towards equal weights (Diebold and Shin, 2019) or restrict them (Zou et al., 2025). With many series, weights are estimated series by series or learned from series features (FFORMA, Montero-Manso et al., 2020); Aiolfi and Timmermann (2006) cluster *forecasting models* by past performance. All of these treat each target series separately or impose one rule on all, and none groups the *target units*.

Decision-focused learning. The loss that matters is the decision cost (Granger, 1969). The data-driven newsvendor literature learns order quantities directly from data (Levi et al., 2015; Ban and Rudin, 2019), prescriptive analytics maps covariates to decisions (Bertsimas and Kallus, 2020), and “smart predict-then-optimize” trains predictors on decision regret (Elmachtoub and Grigas, 2022); see Mandi et al. (2024) for a survey. In energy forecasting, Wang et al. (2019) combine probabilistic load forecasts by minimising pinball loss on the simplex, one series at a time. Most of this work concerns single models or one unit; the exception is data pooling (Gupta and Kallus, 2022), which shrinks the solutions of many small newsvendor-type problems towards a common anchor, which pays even for unrelated problems; decision-focused shrinkage of weights towards pooled weights, one of our benchmarks and the $G = 1$ case of soft GDMA, applies the same idea to forecast combination.

Deep learning for forecasting and for combination weights. Pretrained foundation models such as Chronos (Ansari et al., 2024) forecast unseen series zero-shot with accuracy close to trained models, and deep networks increasingly *produce* combination weights: FFORMA maps series features to weights with gradient boosting (Montero-Manso et al., 2020), and mixture-of-experts model averaging uses a softmax neural gate (Gong et al., 2026). These approaches are flexible, but fit thousands of parameters, give weights that are hard to audit and, except for Gong et al. (2026), lack optimality guarantees for the combined

Table 1: Positioning of GDMA relative to existing approaches.

Approach	Combines candidates	Decision- focused loss	Shares strength across units	Heterogeneous weights
Per-series combination / JMA / forward validation	✓			✓
Pooled combination	✓		✓	
Quantile / general-loss model averaging	✓	✓		✓
Shrinkage towards equal or pooled weights	✓		partial	✓
Data pooling in stochastic optimisation		✓	partial	✓
Feature-based weights (FFORMA)	✓		✓	✓
Neural mixture-of-experts gate	✓	(✓)	✓	✓
Data-driven newsvendor, SPO		✓	✓	
Latent-group panel (quantile) models		(✓)	✓	✓
GDMA (this paper)	✓	✓	✓	✓

decision. We include a foundation model as a candidate and a decision-focused neural gate as a competitor.

Storage siting. Battery siting is usually decided with engineering models of the transmission network, solved by decomposition or stochastic programming (Piansky et al., 2024; Larroux et al., 2026), which take the uncertainty storage must absorb as an input. We make that input endogenous: residual commitment risk depends on how operators combine forecasts, so the value and best location of storage depend on the forecasting policy.

Latent group structures. Panel econometrics models unobserved heterogeneity through latent groups estimated by k -means-type algorithms (Lin and Ng, 2012; Bonhomme and Manresa, 2015; Ando and Bai, 2016) or classifier-Lasso (Su et al., 2016). Gu and Volgushev (2019) extend latent groups to panel quantile regression, the closest statistical relative of GDMA: a grouped quantile regression of demand on the candidate forecasts would be an unconstrained version of our quantile-regression benchmark. These methods group regression parameters, typically fixed effects, for inference on the parameters; GDMA groups simplex weights over existing forecasting systems, with unit-specific critical ratios, and is judged by the out-of-sample cost of its decisions.

The gap. Table 1 summarises the position of GDMA. No existing method simultaneously (a) combines a library of existing candidate rules, including machine-learning and foundation-model forecasts, (b) chooses weights by the downstream decision cost with unit-specific cost asymmetries, and (c) shares strength across many decision units through latent segments whose number is chosen from data, with guarantees on optimality and segment recovery; nor does any link the resulting operational risk to the siting of new storage. The ingredients exist—decision-focused combination, shrinkage across problems, latent groups—and the contribution is their combination, the theory for it, and its use for a planning decision; much of the gain over the best single forecaster is already delivered by decision-focused shrinkage, and grouping adds a smaller refinement. The gap matters most when windows are short, because systems change and old data are stale, and units are many, as in infrastructure planning.

3. Grouped decision-focused model averaging

3.1. The system-wide commitment problem

A planner is responsible for N decision units (in our application the balancing authorities of the U.S. grid). In period t unit i must commit a quantity q_{it} before demand y_{it} is realised. Committing too little costs $u_i > 0$ per unit of shortfall (emergency purchases, reserve activation or, in the limit, curtailed load); committing too much costs $o_i > 0$ per unit of excess (idle capacity that was paid for). The period cost is the newsvendor cost

$$C_i(q, y) = u_i (y - q)^+ + o_i (q - y)^+ = (u_i + o_i) \rho_{\tau_i}(y - q), \quad \tau_i = \frac{u_i}{u_i + o_i}, \quad (1)$$

where $\rho_\tau(r) = r(\tau - \mathbb{K}\{r < 0\})$ is the check function. The system cost that the planner cares about is $\sum_i \sum_t C_i(q_{it}, y_{it})$; under (1) the cost-minimising commitment is the conditional τ_i -quantile of demand (Gneiting, 2011). Critical ratios τ_i differ across units because the consequences of a shortfall differ: an operator that is tightly interconnected with its neighbours can import at moderate cost, while an isolated one cannot.

For every unit a library of M candidate commitment rules is available. Candidate m proposes $q_{it}^{(m)}$, computed from information available before period t ; we write $\mathbf{q}_{it} = (q_{it}^{(1)}, \dots, q_{it}^{(M)})^\top$. The candidates may come from any source: an operator's own published forecast, simple persistence rules, statistical models or machine-learning models, each turned into a commitment by adding its own τ_i -quantile residual margin. The planner combines them linearly,

$$q_{it}(\mathbf{w}) = \mathbf{q}_{it}^\top \mathbf{w}, \quad \mathbf{w} \in \Delta_M = \left\{ \mathbf{w} \in \mathbb{R}^M : w_m \geq 0, \sum_m w_m = 1 \right\}.$$

The simplex constraint keeps the combined commitment inside the range spanned by the candidates, which is what makes the combination interpretable and auditable by operators, and it is known to stabilise estimated weights (Zou et al., 2025).

3.2. Three ways to estimate the weights

Let the estimation window contain periods $t = 1, \dots, T$ for which both candidate commitments and realised demand are observed (a forward-validation sample in the terminology of Zhang and Zhang, 2023). Write $\hat{R}_i(\mathbf{w}) = T^{-1} \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}, y_{it})$ for the empirical cost of unit i .

Unit-specific weights. $\hat{\mathbf{w}}_i = \arg \min_{\mathbf{w} \in \Delta_M} \hat{R}_i(\mathbf{w})$. This respects heterogeneity but estimates $M - 1$ free parameters from T serially dependent observations per unit; it is the decision-focused counterpart of per-series forecast combination, and suffers from the same estimation noise that underlies the forecast-combination puzzle (Smith and Wallis, 2009; Claeskens et al., 2016).

Pooled weights. $\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \sum_i \hat{R}_i(\mathbf{w})$. This uses all NT observations but forces a single policy on units that may need different ones.

Grouped weights (proposed). We posit that units fall into a small number G of latent segments whose members are well served by the same combination. The GDMA estimator solves

$$(\hat{\gamma}, \hat{\mathbf{W}}) = \arg \min_{\gamma \in \{1, \dots, G\}^N, \mathbf{W} = (\mathbf{w}_1, \dots, \mathbf{w}_G) \in \Delta_M^G} \sum_{i=1}^N \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}_{\gamma_i}, y_{it}), \quad (2)$$

Algorithm 1 GDMA for a fixed number of segments G

Require: candidate commitments $\{\mathbf{q}_{it}\}$, demand $\{y_{it}\}$, costs $\{u_i, o_i\}$, G , number of starts S

- 1: compute unit-specific weights $\widehat{\mathbf{w}}_i$ and the regret matrix $D_{ij} = \widehat{R}_i(\widehat{\mathbf{w}}_j) - \widehat{R}_i(\widehat{\mathbf{w}}_i)$
- 2: **for** $s = 1, \dots, S$ **do**
- 3: draw seeds $j_1, \dots, j_G$ by k -means++ on D and set $\mathbf{w}_g \leftarrow \widehat{\mathbf{w}}_{j_g}$
- 4: **repeat**
- 5: $\gamma_i \leftarrow \arg \min_g \widehat{R}_i(\mathbf{w}_g)$ for all i $\triangleright$ assignment step
- 6: re-seed any empty segment with the unit of largest regret
- 7: $\mathbf{w}_g \leftarrow \arg \min_{\mathbf{w} \in \Delta_M} \sum_{i: \gamma_i = g} \widehat{R}_i(\mathbf{w})$ for all g $\triangleright$ pooled convex step
- 8: **until** memberships do not change
- 9: **end for**
- 10: **return** the solution with the lowest objective (2)

and unit i commits $\mathbf{q}_{i,T+1}^\top \widehat{\mathbf{w}}_{\gamma_i}$. $G = 1$ gives pooled weights and $G = N$ gives unit-specific weights, so the number of segments indexes the whole range between the two extremes. Three features separate (2) from existing proposals. First, the criterion is the decision cost in the planner’s own units (money or energy), so units are grouped by *how they should act*, not by how their forecasts look. Second, memberships and weights are estimated jointly; a two-step procedure that clusters noisy unit-specific weights and then re-estimates is a natural but inferior alternative (Section 5). Third, because each unit keeps its own critical ratio and its own candidate margins, units in one segment share a combination rule but not a commitment level.

3.3. Computation

Problem (2) is a mixed-integer problem, but it has the structure of k -means with a non-Euclidean, data-dependent distortion, as in the grouped fixed-effects estimator of Bonhomme and Manresa (2015). Given memberships, the weights of each segment solve a pooled convex problem on the simplex; given weights, each unit joins the segment whose weights give it the lowest in-window cost. Algorithm 1 alternates the two steps. The weight step is a linear program; for speed we minimise a softplus-smoothed check function, $\rho_r^\varepsilon(r) = \tau r + \varepsilon \log(1 + e^{-r/\varepsilon})$, with bandwidth ε equal to 2% of the unit’s typical decision error, by sequential quadratic programming. On our data the smoothed solution is within 0.01% of the exact linear-programming optimum in cost. Each iteration weakly decreases the objective, so the algorithm converges to a local solution in finitely many steps; we run several starts and keep the best. Starting values are drawn by a k -means++ rule (Arthur and Vassilvitskii, 2007) in *regret space*: the next seed unit is drawn with probability proportional to the extra cost that units would incur if forced to use the closest existing seed’s own weights. In rolling applications the memberships of the previous window provide an additional warm start.

3.4. Soft segments

Segments need not be exact. When units within a segment are similar but not identical, a unit may benefit from a rule that sits between its own estimated weights and those of its segment. *Soft GDMA* uses

$$\widehat{\mathbf{w}}_i^{\text{soft}} = (1 - \kappa) \widehat{\mathbf{w}}_i + \kappa \widehat{\mathbf{w}}_{\gamma_i}, \quad \kappa \in [0, 1], \quad (3)$$

where $\widehat{\mathbf{w}}_i$ are the unit-specific weights and $\widehat{\mathbf{w}}_{\gamma_i}$ the GDMA weights of the unit's segment. Because the simplex is convex, $\widehat{\mathbf{w}}_i^{\text{soft}} \in \Delta_M$. The family nests every benchmark: $\kappa = 1$ is GDMA, $\kappa = 0$ is unit-specific weights, $G = 1$ is shrinkage towards pooled weights (a decision-focused analogue of [Diebold and Shin, 2019](#)), and $G = 1, \kappa = 1$ is pooled weights. The number of segments is chosen exactly as for GDMA (next subsection); given G , κ is chosen on the same hold-out block from $\{0, 0.25, 0.5, 0.75, 1\}$. Choosing G first keeps the search small, which matters because hold-out costs are noisy when windows are short. Segment centres play the role of data-driven prior means, in the spirit of empirical Bayes shrinkage towards group-specific targets.

3.5. Choosing the number of segments

We split the estimation window into an early block (the first two thirds) and a late block. For each $G \in \{1, \dots, G_{\max}\}$ we estimate (2) on the early block and record every unit's cost on the late block under its estimated membership and weights. Let $G^\dagger$ minimise the total late-block cost. Following the one-standard-error rule ([Hastie et al., 2009](#)), we choose the smallest G whose total late-block cost exceeds that of $G^\dagger$ by no more than one standard error of the summed unit-level cost differences, and then re-estimate on the full window. The rule respects time order and prefers parsimonious segmentations, which are easier to explain to the operators who would have to adopt them.

3.6. From commitment rules to storage siting

The combination rule is an operational decision, re-estimated every week. A planner also faces a slower, capital-intensive decision: where to install new flexible capacity. We link the two with a two-stage design. In the operational stage, the learned rule leaves each unit with hourly shortfalls $s_{it} = (y_{it} - q_{it})^+$ and surpluses $x_{it} = (q_{it} - y_{it})^+$. In the planning stage, a battery with power P_i (MW) and energy $\bar{h}P_i$ (MWh) installed in unit i discharges to cover shortfalls and recharges from committed-but-unused capacity, which has already been paid for. With state of charge e_{it} and round-trip efficiency η , the chronological dispatch is

$$\begin{aligned} d_{it} &= \min\{P_i, s_{it}, e_{it}\}, & c_{it} &= \min\{P_i, x_{it}, (\bar{h}P_i - e_{it} + d_{it})/\eta\}, \\ e_{i,t+1} &= e_{it} - d_{it} + \eta c_{it}, \end{aligned}$$

and every MWh discharged avoids the underage cost u_i . The annual value of battery power, $V_i(P)$, equal to u_i times the expected energy discharged per year, is non-decreasing and typically concave in P , because each additional MW of power is called on only in rarer and larger shortfalls; where the estimated curve is not concave on the grid we replace later increments by the running minimum of the marginal values, which yields a concave lower bound. Given an annualised cost κ per MW of battery power, the planner solves

$$\max_{P_1, \dots, P_N \geq 0} \sum_{i=1}^N \{V_i(P_i) - \kappa P_i\} \quad \text{subject to} \quad \sum_{i=1}^N P_i \leq B, \quad (4)$$

where B is a storage budget or deployment target (for example a national or continental procurement programme). Because each (concavified) V_i is concave, a greedy algorithm that adds power in small increments to the unit with the highest marginal value is optimal for (4); without the budget constraint, each unit simply installs power until its marginal value falls to κ . We estimate V_i on a grid of battery sizes from the out-of-sample shortfall and surplus paths of each commitment rule.

The two stages interact in two ways that matter for management. First, better commitment rules shrink the shortfalls that storage must cover, so part of the storage requirement can be met by better forecast combination instead of new hardware; we measure this *storage equivalent* of better forecasting as the battery power that the status-quo rule needs to reach the cost of the better rule without storage. Second, the siting decision depends on the rule: batteries placed where the status-quo rule is weakest may be placed where they add little once operations improve.

3.7. Fairness across decision units

At continental scale the objective in (2) has a distributional consequence. Because costs are measured in money or energy, the largest systems dominate the criterion, and a segment rule that is excellent for a large system may be mediocre for a small or data-poor one that happens to share its segment. The same holds for storage: problem (4) places batteries where the value is highest in absolute terms. Energy-justice scholarship stresses that the distribution of the benefits of energy infrastructure matters in its own right (Sovacool and Dworkin, 2015), and the operations literature measures the efficiency cost of fair allocations by the *price of fairness* (Bertsimas et al., 2011). We therefore add two fair variants and audit every method for how its benefits are distributed.

Fair soft GDMA (one area, one vote). Let c_i^0 be the in-window cost of unit i under the reference forecaster (candidate 1, the official forecast where one exists). Fair soft GDMA solves the soft GDMA problem with every unit’s costs divided by c_i^0 (normalised to mean one), that is, with u_i and o_i replaced by $\omega_i u_i$ and $\omega_i o_i$, $\omega_i \propto 1/c_i^0$. Each unit then counts in proportion to its *relative* improvement, so that a 10% saving in a 70 MW municipal utility weighs as much as a 10% saving in a 90 GW regional market. This is the relative-cost analogue of group-robust learning (Sagawa et al., 2020); it changes only the weights of the criterion, not the algorithm.

Fair storage siting. We consider two fair versions of (4) over continents c (any grouping, such as countries or income groups, works the same way). *Proportional floors* require every continent to receive at least a share λ of its share of demand in the budget; the floors are filled greedily within each continent and the rest of the budget greedily across all units. *Maximin siting* allocates each increment of battery power to the continent whose relative benefit—the value captured divided by its commitment cost without storage—is currently the lowest, and within it to the unit with the highest marginal value, which implements a max–min notion of fairness (Bertsimas et al., 2011). The price of fairness is the share of the efficient programme’s value that each fair version gives up.

Audit. For every method we report the distribution across units of the ratio of its cost to the unit’s cost under the reference forecaster, the share of units made worse off than under the reference forecaster and than under their own best single candidate (chosen on the estimation window), the median ratio of the worst-served continent, and separate ratios for data-rich and data-poor systems.

4. Theoretical properties

This section studies the global minimiser of (2) for a given G . We show that (i) its expected decision cost is close to that of the best grouped policy, with an estimation error that separates the price of learning memberships from the price of learning weights; (ii) it is asymptotically optimal and, under a latent segment structure, attains the risk

of the infeasible unit-specific oracle; and (iii) it recovers the segments with probability approaching one, in which case its weights coincide with those of an estimator that is told the segments. Proofs are in Appendix A. The scope of these results should be stated plainly. They concern the exact global minimiser of the grouped criterion for a fixed number of segments, under stationarity, bounded losses and a high-level concentration condition; the estimator used in the applications is computed by a local search, with G and the shrinkage intensity chosen on a hold-out, and the data are neither stationary nor independent across units. The theory therefore explains why grouping can beat both unit-specific and pooled weights and what it costs to learn the segments; it does not deliver numerical guarantees for the empirical estimator.

4.1. Setting and assumptions

Let $\mathbf{z}_{it} = (y_{it}, \mathbf{q}_{it})$. Define the population risk of unit i , $R_i(\mathbf{w}) = \mathbb{E} C_i(\mathbf{q}_{it}^\top \mathbf{w}, y_{it})$, and for a grouped policy $(\gamma, \mathbf{W})$ the average risk and its empirical analogue

$$R(\gamma, \mathbf{W}) = \frac{1}{N} \sum_{i=1}^N R_i(\mathbf{w}_{\gamma_i}), \quad \hat{R}(\gamma, \mathbf{W}) = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T C_i(\mathbf{q}_{it}^\top \mathbf{w}_{\gamma_i}, y_{it}).$$

Let $\mathcal{P}_G = \{1, \dots, G\}^N \times \Delta_M^G$ and $R_G^* = \inf_{\mathcal{P}_G} R$.

Assumption 1 (Stationarity and boundedness). *For each i , $\{\mathbf{z}_{it}\}_t$ is strictly stationary, $|y_{it}| \leq B$ and $\|\mathbf{q}_{it}\|_\infty \leq B$ almost surely, and $\max_i(u_i + o_i) \leq \bar{a}$.*

Stationarity refers to the candidate commitments as well as demand; it holds, for instance, when every candidate is computed from a rolling window of fixed length of a stationary process. Boundedness is natural once demand is normalised by a unit's scale. Under Assumption 1 each loss lies in $[0, L]$ with $L = 2\bar{a}B$ and is Lipschitz in the weights: $|C_i(\mathbf{q}^\top \mathbf{w}, y) - C_i(\mathbf{q}^\top \mathbf{w}', y)| \leq \bar{a}B \|\mathbf{w} - \mathbf{w}'\|_1$.

Assumption 2 (Concentration). *There are constants $c > 0$ and an effective sample length $T_e \leq T$ such that, for every fixed $(\gamma, \mathbf{W}) \in \mathcal{P}_G$ and every $x > 0$,*

$$\Pr(|\hat{R}(\gamma, \mathbf{W}) - R(\gamma, \mathbf{W})| > x) \leq 2 \exp(-cNT_e x^2/L^2),$$

and, for every i and fixed $\mathbf{w}$, $\Pr(|\hat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| > x) \leq 2 \exp(-cT_e x^2/L^2)$.

Assumption 2 is a high-level condition. It holds with $T_e = T$ when observations are independent (Hoeffding's inequality) and, when units are independent and each $\{\mathbf{z}_{it}\}_t$ is geometrically β -mixing, with $T_e = T/(C \log T)$ by the blocking argument of Yu (1994) or the Bernstein inequality of Merlevède et al. (2011). Cross-sectional dependence through common shocks (weather, in our application) reduces the effective number of independent units; the results continue to hold with N replaced by an effective cross-sectional size.

4.2. An oracle inequality

Theorem 1 (Oracle inequality). *Let Assumptions 1–2 hold and let $(\hat{\gamma}, \widehat{\mathbf{W}})$ be a global minimiser of (2). For any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$R(\hat{\gamma}, \widehat{\mathbf{W}}) \leq R_G^* + 2L \sqrt{\frac{GM \log(3NT_e) + N \log G + \log(2/\delta)}{cNT_e}} + \frac{4\bar{a}B}{NT_e}. \quad (5)$$

The square-root term has a transparent decomposition. Per unit, learning the segment membership costs $\log G/T_e$, while learning the segment weights costs $GM \log(NT_e)/(NT_e)$. The same argument applied to unit-specific weights ($G = N$) gives $M \log(NT_e)/T_e$, and applied to pooled weights ($G = 1$) gives $M \log(NT_e)/(NT_e)$ but a larger approximation error $R_1^* - R_N^*$. Grouping therefore replaces the M per-unit parameters of unit-specific weights by a single discrete label that costs $\log G$ rather than $M \log(NT_e)$ to learn. This is the formal sense in which GDMA resolves the tension between the noise of unit-specific weights and the bias of pooled weights: with $M = 7$ candidates, $N = 46$ units, $G = 4$ segments and an effective length of $T_e = 28$ (one observation per day of a four-week window), the variance proxy under the square root is 0.23 for GDMA against 2.07 for unit-specific weights, a factor of three in the square-root term, while the approximation error of GDMA is zero whenever the segment structure is correct. With unknown constants L and c both bounds are loose at these sizes, so the comparison concerns how the two error terms scale, not the size of the bounds.

4.3. Asymptotic optimality

Following the model-averaging literature (Hansen, 2007; Hansen and Racine, 2012; Zhang and Liu, 2023), call a weight choice asymptotically optimal if the ratio of its risk to the lowest attainable risk in its class tends to one.

Corollary 1 (Asymptotic optimality). *Let the assumptions of Theorem 1 hold, with G and M possibly growing, and suppose $R_G^* \geq \kappa > 0$. If $GM \log(NT_e)/(NT_e) \rightarrow 0$ and $\log G/T_e \rightarrow 0$, then $R(\hat{\gamma}, \widehat{\mathbf{W}})/R_G^* \rightarrow_p 1$. If, in addition, there are memberships γ^0 with $G_0 \leq G$ segments such that $\arg \min_{\mathbf{w}} R_i(\mathbf{w})$ depends on i only through γ_i^0 , then $R_G^* = R_N^* = N^{-1} \sum_i \min_{\mathbf{w}} R_i(\mathbf{w})$ and*

$$\frac{R(\hat{\gamma}, \widehat{\mathbf{W}})}{N^{-1} \sum_i \min_{\mathbf{w} \in \Delta_M} R_i(\mathbf{w})} \rightarrow_p 1.$$

The second statement is the relevant one for practice: under a segment structure, GDMA attains the risk of the infeasible benchmark in which every unit knows its own optimal weights, although it estimates only $G(M-1)$ continuous parameters. Unit-specific weights attain the same limit only if $M \log(NT_e)/T_e \rightarrow 0$, a much stronger requirement when windows are short because the system is non-stationary and old data are of limited use. The condition $R_G^* \geq \kappa$ excludes the degenerate case of perfect foresight, which never occurs in commitment problems.

4.4. Recovery of the segments

Assumption 3 (Segment structure). (a) *There are $\gamma^0 \in \{1, \dots, G\}^N$ and distinct $\mathbf{w}_1^0, \dots, \mathbf{w}_G^0 \in \Delta_M$ such that, for all i and all $\mathbf{w} \in \Delta_M$, $R_i(\mathbf{w}) - R_i(\mathbf{w}_{\gamma_i^0}^0) \geq \lambda \|\mathbf{w} - \mathbf{w}_{\gamma_i^0}^0\|_2^2$ for some $\lambda > 0$. (b) $\min_{g \neq h} \|\mathbf{w}_g^0 - \mathbf{w}_h^0\|_2 \geq d > 0$ and every segment contains at least πN units, $\pi > 0$.*

Part (a) is a quadratic-growth condition. It holds when the conditional density of demand given the candidates is bounded away from zero near the optimal commitment and the candidates are not collinear, because the Hessian of R_i is then $(u_i + o_i) \mathbb{E}[f_{y|q}(\mathbf{q}^\top \mathbf{w}) \mathbf{q} \mathbf{q}^\top]$, which is positive definite on the tangent space of the simplex. Part (b) requires segments to be genuinely different and non-negligible, as in Bonhomme and Manresa (2015).

Theorem 2 (Segment recovery and oracle equivalence). *Let Assumptions 1–3 hold with G equal to the true number of segments, and let $N, T_e \rightarrow \infty$ with $\log N/T_e \rightarrow 0$. Then, up to a relabelling of segments, (i) $\max_g \|\widehat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \rightarrow_p 0$; (ii) $\Pr(\widehat{\boldsymbol{\gamma}} \neq \boldsymbol{\gamma}^0) \leq N C_M \exp(-c' T_e) + o(1) \rightarrow 0$ for constants $C_M, c' > 0$ that do not depend on N or T ; (iii) with probability approaching one, $\widehat{\mathbf{W}}$ equals the infeasible estimator $\widetilde{\mathbf{W}} = \arg \min_{\mathbf{W}} \widehat{R}(\boldsymbol{\gamma}^0, \mathbf{W})$ that knows the segments.*

Part (iii) is an exact equivalence, not an asymptotic approximation: on the event that memberships are recovered, the joint minimiser and the known-segment minimiser are the same object. Any distributional result available for the known-segment estimator therefore transfers to GDMA without modification. The recovery condition $\log N/T_e \rightarrow 0$ is mild: the number of units may grow polynomially in the window length.

Soft GDMA. For fixed (G, κ) , R_i is convex in $\mathbf{w}$, so the risk of (3) satisfies $R_i(\widehat{\mathbf{w}}_i^{\text{soft}}) \leq (1 - \kappa)R_i(\widehat{\mathbf{w}}_i) + \kappa R_i(\widehat{\mathbf{w}}_{\widehat{\gamma}_i})$. Averaging over units, the excess risk of soft GDMA is at most the κ -weighted average of the excess risks of unit-specific weights and of GDMA, each controlled by Theorem 1. Under a segment structure, therefore, any κ bounded away from zero inherits the faster rate of GDMA for the κ share of the combination, while $\kappa < 1$ buys robustness when segments are only approximate. These statements hold for fixed (G, κ) . We do not analyse the data-dependent choice of (G, κ) on the hold-out, nor why soft GDMA with an estimated κ can outperform both of its limits; a formal treatment of the hold-out rule under serial and cross-sectional dependence is left for future work.

Weighted and fair criteria. All results of this section hold for criteria in which unit i 's costs are multiplied by a weight $\omega_i \in [\underline{\omega}, \bar{\omega}]$ with $0 < \underline{\omega} \leq \bar{\omega} < \infty$: the weights only rescale u_i and o_i , so Assumptions 1–3 hold with $\bar{a}$ replaced by $\bar{\omega}\bar{a}$ and λ by $\omega\lambda$. Fair soft GDMA uses data-dependent weights $\omega_i \propto 1/c_i^0$; treating them as fixed within a window, Theorem 1 bounds its weighted excess risk. Under the segment structure of Corollary 1, the weighted and unweighted criteria have the same minimiser, because every unit attains its own optimal weights, and since each unit's excess risk is non-negative the unweighted excess risk is at most $1/\underline{\omega}$ times the weighted one (weights normalised to mean one). Fairness then costs nothing asymptotically, although the factor $1/\underline{\omega}$ can be large when the areas' costs differ by orders of magnitude. When segments are only approximate, the fair criterion trades some total cost for a more even distribution of the gains; this is the price of fairness that we measure empirically.

5. Monte Carlo evidence

5.1. Design

The simulation examines the finite-sample behaviour of the estimator as it is used in the applications, with G and κ chosen on a hold-out; it does not verify the rates of Section 4, which concern the exact minimiser for a fixed G . It is deliberately simpler than the applications: units are independent, the data are stationary, and costs are linear, so it isolates the trade-off between the noise of unit-specific weights and the bias of pooled weights.

We simulate $N \in \{50, 200\}$ units and $M = 5$ candidate commitment rules. Unit i has demand $y_{it} = \mu_i + x_{it}$, with $\mu_i \sim U[20, 100]$ and x_{it} a stationary AR(1) process with coefficient 0.5 and standard deviation 5, and critical ratio $\tau_i \sim U[0.6, 0.95]$ ($u_i = \tau_i/(1 - \tau_i)$, $o_i = 1$). Candidate m forecasts $f_{itm} = y_{it} - e_{itm}$, where the errors follow AR(1)

Table 2: Monte Carlo results: out-of-sample newsvendor cost relative to the unit-specific oracle (means over 50 replications), average selected number of segments $\bar{G}$ of GDMA, and adjusted Rand index (ARI; Hubert and Arabie, 1985) between estimated and true segments for GDMA and two-step clustering. Bold: lowest in the row. Lower is better. The ratio cannot fall below 1; ARI: higher is better.

Design	N	T	Out-of-sample cost / oracle							$\bar{G}$	ARI	
			EW	Pooled	Per-unit	Shrink	2-step	GDMA	Soft		GDMA	2-step
Grouped	50	48	1.496	1.414	1.084	1.090	1.055	1.053	1.041	3.0	0.90	0.75
	50	168	1.491	1.410	1.033	1.033	1.031	1.031	1.023	3.8	0.88	0.79
	50	672	1.481	1.398	1.013	1.013	1.019	1.018	1.012	5.3	0.68	0.67
	200	48	1.491	1.418	1.086	1.089	1.049	1.040	1.035	3.1	0.94	0.79
	200	168	1.492	1.419	1.032	1.032	1.028	1.024	1.020	4.6	0.78	0.73
	200	672	1.491	1.416	1.012	1.012	1.021	1.017	1.011	5.9	0.61	0.66
Homogeneous	50	48	1.321	1.027	1.097	1.027	1.032	1.027	1.027	1.0	—	—
	50	168	1.322	1.025	1.036	1.020	1.026	1.024	1.020	1.2	—	—
	50	672	1.322	1.025	1.015	1.013	1.017	1.016	1.012	2.4	—	—
	200	48	1.321	1.026	1.094	1.025	1.027	1.026	1.026	1.0	—	—
	200	168	1.320	1.025	1.038	1.020	1.026	1.024	1.020	1.2	—	—
	200	672	1.321	1.025	1.014	1.013	1.017	1.015	1.012	2.9	—	—
Continuous	50	48	1.206	1.148	1.091	1.075	1.090	1.097	1.069	2.3	—	—
	50	168	1.206	1.148	1.035	1.036	1.051	1.051	1.030	4.4	—	—
	50	672	1.203	1.145	1.014	1.014	1.035	1.035	1.014	5.3	—	—
	200	48	1.202	1.146	1.094	1.075	1.090	1.085	1.062	2.5	—	—
	200	168	1.205	1.149	1.035	1.038	1.050	1.050	1.030	4.7	—	—
	200	672	1.206	1.150	1.013	1.013	1.040	1.036	1.014	6.0	—	—

processes with coefficient 0.5 and innovations with unit-specific scales s_{im} and skewness signs $k_{im} \in \{-1, 0, +1\}$ (a centred gamma distribution for ± 1 , a normal distribution for 0). Each candidate becomes a commitment by adding the τ_i -quantile of its errors in a pre-sample of 200 periods. Three designs define the heterogeneity: *grouped*—every unit belongs to one of three segments with segment-specific error profiles (s_i, k_i), so that different candidates are best in different segments; *homogeneous*—all units share the first profile; *continuous*—each unit’s error scales are a Dirichlet(1,1,1) mixture of the three profiles, so that no two units share their optimal weights and GDMA’s segment structure is misspecified. Weights are estimated on $T \in \{48, 168, 672\}$ periods and evaluated on 1,000 new periods. We report the out-of-sample cost relative to an infeasible oracle that uses, for every unit, the exact cost-minimising weights on the evaluation sample itself. Each cell uses 50 replications.

5.2. Results

Table 2 supports the theory and clarifies when grouping pays.

Soft GDMA is the most reliable method. It has the lowest mean cost, or ties for it at three decimals, in every one of the 18 cells. Its advantage is largest when windows are short: with true segments and $T = 48$ it is 3.5–4.1% above the oracle against 8.4–8.6% for unit-specific weights, and even under continuous heterogeneity, where segments are misspecified, it beats unit-specific weights (6.2–6.9% against 9.1–9.4% above the oracle) and shrinkage towards pooled weights (7.5%). When units share one error profile it matches pooled weights for short windows and beats them for long ones, where the heterogeneity in critical ratios makes some unit-level adjustment worthwhile.

Hard segments trade bias for variance exactly as Theorem 1 predicts. Plain GDMA beats unit-specific weights in 90–100% of the replications with true segments and $T = 48$, and in all replications with homogeneous units for $T \leq 168$. With long windows ($T = 672$), unit-specific weights are estimated precisely and become slightly better (1.2–1.3% against 1.7–1.8% above the oracle on average; GDMA is better in at most 2% of the replications). Here GDMA selects five to six segments on average rather than three, because units within a true segment have different critical ratios and therefore different optimal weights: with enough data the hold-out rule correctly detects this finer structure, which is also why the ARI falls from about 0.9 at $T = 48$ to about 0.65 at $T = 672$. Soft GDMA removes this residual bias through unit-level shrinkage and its mean cost is never above that of unit-specific weights.

Joint estimation recovers segments better than clustering estimated weights. With $T = 48$, the ARI of GDMA is 0.90–0.94, against 0.75–0.79 for two-step k -means on noisy unit-level weights, and GDMA’s mean cost is lower than, or equal to, that of two-step clustering in every grouped cell with $T \leq 168$. When units share one error profile, the one-standard-error rule selects a single segment in most replications with $T \leq 168$ ($\bar{G} = 1.0$ –1.2).

6. Day-ahead capacity commitment on the U.S. grid

6.1. Data

Reading convention. Every cost in this paper is an operating cost, so *lower is better*; a ratio below 1 means a lower cost than the benchmark named in the table, and “more expensive” means worse. Gains, savings, storage values and the adjusted Rand index are the opposite: higher is better. Each caption states the direction.

We use the public U.S. Hourly Electric Grid Monitor (Form EIA-930; [U.S. Energy Information Administration, 2026](#)), downloaded on 29 September 2026: hourly demand, day-ahead demand forecast and total interchange for each balancing authority (BA) in the lower 48 states, 1 January 2019 to 26 September 2026 in UTC hours, excluding the fourteen regional aggregates. Following [Ruggles et al. \(2020\)](#), non-positive values and those outside $[0.3, 3]$ times the preceding seven-day median are treated as missing (one-sided filter, applied to the official forecast against its own history) and gaps of up to three hours are interpolated. We keep BAs with at least 97% valid demand, at least 93% valid official forecasts, mean demand above 50 MW and a median absolute official forecast error below 15% of median demand. Forty-six BAs remain (Table C.20 in Appendix [Appendix C](#)); they range from PJM (mean demand 92.5 GW) to the City of Homestead (71 MW), include all seven independent system operators and cover 97% of lower-48 demand. Ten mostly small BAs are dropped; 0.32% of retained BA-hours are missing, and 0.6% of official forecasts are replaced by the mean of the other candidates.

6.2. Decision problem and cost calibration

Before the day-ahead gate closure on day $d - 1$, BA i commits hourly capacity $q_{i,d,h}$, $h = 0, \dots, 23$, for the (UTC) operating day d . Demand is observed only to the end of day $d - 2$, since UTC day $d - 1$ ends after any U.S. day-ahead deadline; the information set is demand and rule costs up to day $d - 2$ plus the published forecast for day d . This is conservative: using hours of day $d - 1$ would slightly favour the data-driven candidates. Excess commitment costs $o_i = 1$ per MWh (the normalisation); shortfalls cost $u_i = \tau_i / (1 - \tau_i)$, where the critical ratio τ_i reflects how easily the BA can lean on

its neighbours. As a proxy we use the 2019 range between the 5th and 95th percentiles of hourly interchange divided by mean demand, and let τ_i decline linearly in its rank from 0.95 for the least variable BA to 0.80 for the most. Interchange variability is not transfer capability, so the proxy only orders BAs; Section 6.7 reports common critical ratios from 0.5 to 0.95 (Section 7: 0.99 for the global panel). Almost islanded ERCOT receives $\tau = 0.95$ ($u_i = 19$); Chelan County PUD, in the interconnected Pacific Northwest, receives $\tau = 0.80$ ($u_i = 4$).

6.3. Candidate rules

Seven candidate forecasts are produced for every BA, day and hour: (1) the official forecast; (2) persistence (same hour of $d - 2$); (3) weekly persistence (same hour of $d - 7$); (4) the mean of the same hour over $d - 2, \dots, d - 8$; (5) the mean of the same hour on $d - 7, d - 14$ and $d - 21$; (6) a BA-and-hour-specific linear regression on the $d - 2$ and $d - 7$ values, the official forecast and weekday dummies, on a rolling 56-day window refitted weekly; and (7) a global LightGBM model (Ke et al., 2017) trained on lags, calendar variables and the official forecast, all scaled by the BA’s recent mean demand, refitted every 28 days on the preceding 365 days. Each forecast $f^{(m)}$ becomes a candidate commitment with its own safety margin,

$$q_{i,d,h}^{(m)} = f_{i,d,h}^{(m)} \left\{ 1 + \widehat{Q}_{\tau_i}(e_i^{(m)}) \right\},$$

where $\widehat{Q}_{\tau_i}(e_i^{(m)})$ is the τ_i -quantile of the relative errors $(y - f^{(m)}) / \max\{f^{(m)}, 0.2\bar{y}_{i,t}\}$ of forecast m over the 28 days ending at $d - 2$; the 20% floor on the BA’s mean demand $\bar{y}_{i,t}$ over the week before day t bounds the margin. Given the 28-day margin window, the 56-day window cap and the two-day lag, candidates exist from February 2020 and evaluation runs from 1 May 2020 to 26 September 2026: 2,340 days, 335 weekly re-estimations and 25.8 billion MWh of demand.

6.4. Competing methods

Every seven days, at origin d_0 , each method re-estimates its combination from the W most recent days with known outcomes, $d_0 - W - 1, \dots, d_0 - 2$, $W \in \{7, 14, 28, 56\}$ (24 hourly observations per BA per day), and commits for the next seven days. Besides GDMA and soft GDMA ($G_{\max} = 6$) we consider the official-forecast policy (candidate 1 alone), the reference rule, a stylisation of actual commitment, which runs through market clearing; equal weights over the seven candidates; the best single candidate per BA; pooled and per-BA decision-focused weights; shrinkage of per-BA towards pooled weights (cf. Diebold and Shin, 2019); two-step clustering, which runs k -means on per-BA weight vectors and re-estimates pooled weights within clusters; a quantile-regression combination of the seven forecasts at each BA’s τ_i without the simplex constraint (Wang et al., 2019); and two *forecast-then-commit* methods, which estimate per-BA or grouped weights by minimising scaled squared forecast error and then add the τ_i -quantile of the combined forecast’s relative errors over 28 days, isolating the value of estimating for the decision rather than for accuracy. The number of clusters, G , κ and the shrinkage intensity are chosen on one hold-out split in which every third day, counted back from the window’s end, is a validation day.

6.5. Results

Table 3 gives the main result.

Table 3: Out-of-sample system-wide commitment cost, 1 May 2020 to 26 September 2026, relative to committing on the official day-ahead forecasts. W is the estimation window length. Bold: lowest cost in the column. Stars: Diebold–Mariano test of equal daily system cost against soft GDMA with Newey–West variance (7 lags); *, **, ***: $p < 0.10, 0.05, 0.01$. Lower is better: values below 1 mean a lower cost than the benchmark.

Method	System cost relative to the official-forecast policy				Share of BAs improved ^a
	$W = 7$ days	$W = 14$ days	$W = 28$ days	$W = 56$ days	
Official forecast	1.0000***	1.0000***	1.0000***	1.0000***	0.00
Equal weights	1.6901***	1.6902***	1.6902***	1.6902***	0.09
Best single (per BA)	0.8942***	0.8692***	0.8585***	0.8499***	0.50
Pooled DF weights	0.8556***	0.8492	0.8446***	0.8400***	0.52
Per-BA DF weights	0.8898***	0.8581***	0.8422***	0.8308***	0.54
Shrinkage to pooled	0.8661	0.8513**	0.8385***	0.8297**	0.54
Two-step k -means	0.8589***	0.8446**	0.8348†	0.8286	0.57
Forecast-then-commit, per BA	0.9300***	0.8929***	0.8673***	0.8480***	0.50
Forecast-then-commit, grouped	0.8824***	0.8716***	0.8548***	0.8471***	0.52
GDMA	0.8642**	0.8485	0.8356†	0.8300	0.59
Quantile-regression comb.	1.2970***	1.0186***	0.8834***	0.8455***	0.52
GDMA, minimum hold-out rule	0.8735***	0.8506	0.8383***	0.8300*	0.59
Soft GDMA	0.8673	0.8492	0.8355†	0.8288	0.63

^a Share of the 46 BAs whose own cost is lower than under the official-forecast policy ($W = 28$).

First, deciding for the decision pays. Soft GDMA lowers the system-wide commitment cost by 13–17% relative to the official forecasts (significant at the 1% level for every window), and every decision-focused simplex combination by 11–17%. The forecast-then-commit methods save 7–15% and are significantly more expensive than soft GDMA for every window, most at $W = 7$ (0.930 and 0.882 against 0.867), as accuracy-optimal weights ignore the asymmetric costs. The unconstrained quantile-regression combination is competitive only with long windows (0.846 at $W = 56$) and poor with short ones (1.297 at $W = 7$), where dropping the simplex costs more in variance than it gains. Equal weights are worst, raising cost by 69%.

Second, neither extreme is right. Per-BA weights are costly with short windows (0.890 at $W = 7$, 4.0% above pooled weights), pooled weights with long windows (0.840 at $W = 56$, 1.4% above the best grouped rule), and their ranking reverses between $W = 14$ and $W = 28$, the bias–variance trade-off behind GDMA.

Third, grouping helps, but the three grouped rules are statistically indistinguishable. For $W \geq 14$ the three cheapest methods are two-step clustering, GDMA and soft GDMA, and at $W = 28$ they form the 90% model confidence set of Hansen et al. (2011) (marked † in Table 3), which excludes shrinkage (MCS p -value 0.03) and all other rules. They differ by less than 0.2% at $W = 28$ and $W = 56$ and about 0.5% at $W = 14$; two-step clustering is marginally cheapest, and at $W = 7$ pooled weights are cheapest (0.856, against 0.859–0.867 for the grouped rules). Soft GDMA beats the official forecast for the largest share of BAs (63%, against 57–59% for the other grouped rules and 54% for shrinkage), and the one-standard-error rule lowers GDMA’s cost at $W = 7$ (0.864 against 0.874 for the hold-out minimum). Which grouped variant is used matters less than using cross-sectional structure.

Fourth, the savings are economically meaningful. At $W = 28$ the official-forecast policy costs 0.0779 units per MWh of demand and soft GDMA 0.0651, a saving of 0.0128.

Table 4: Secondary metrics for $W = 28$ days. Lower is better for cost; the shortfall frequency should be close to the target. DM statistic: positive values mean the method is more expensive than soft GDMA.

Method	Scale-free cost ^b (rel. to EW)	Shortfall frequency	DM statistic vs soft GDMA ^c
Official forecast	0.6967	0.137	23.68
Equal weights	1.0000	0.089	36.71
Best single (per BA)	0.6131	0.142	7.46
Pooled DF weights	0.6122	0.120	3.38
Per-BA DF weights	0.6030	0.121	8.17
Shrinkage to pooled	0.6008	0.120	4.32
Two-step k -means	0.6035	0.124	-0.39
Forecast-then-commit, per BA	0.6146	0.149	5.47
Forecast-then-commit, grouped	0.6115	0.143	4.18
GDMA	0.6026	0.125	0.11
Quantile-regression comb.	0.6166	0.159	7.84
GDMA, minimum hold-out rule	0.6007	0.125	2.95
Soft GDMA	0.5984	0.121	—
Target (mean τ_i across BAs)		0.125	

^b Each BA’s cost divided by its mean hourly demand, averaged over BAs, relative to equal weights. ^c Diebold–Mariano statistic; positive values: the method has higher daily system cost than soft GDMA.

With roughly 4.0 billion MWh of annual demand (25.8 billion over the evaluation period) and idle capacity valued at an illustrative \$10 per MWh, this is roughly \$500 million a year under the stylised cost model. The saving is not universal: 63% of BAs are better off than under their own official forecast (last column of Table 3), and the official forecast is better for the rest.

Table 4 and Figure 3 confirm this: with every BA weighted equally soft GDMA is still cheapest, so the result is not driven by the three largest ISOs, and the saving builds up steadily. The decision-focused simplex methods hit the target shortfall frequency of 12.5% (the average of $1 - \tau_i$) closely (12.0–12.5%); the official-forecast policy, the forecast-then-commit methods and the quantile-regression combination fall short too often (13.7–15.9%).

6.6. Segments

GDMA uses one segment in 16% of the weeks and two to four in two thirds of them (Figure 4, left); soft GDMA’s average shrinkage towards the segment rule is only $\kappa = 0.14$, so it mostly refines unit-level weights. The co-membership matrix (right) shows two stable segments. The first has 16 BAs, among them PJM and CAISO, and weights the hourly regression 0.38, LightGBM 0.39 and the official forecast 0.20. The second has 29 BAs, including MISO and ERCOT, and puts 0.44 on the official forecast; the Los Angeles Department of Water and Power often forms its own segment. The segments have similar average critical ratios (0.886 and 0.868) and do not follow geography or market design; both mix ISOs and vertically integrated utilities from the East and West. They differ in how much the operator’s forecast adds to what recent demand reveals.

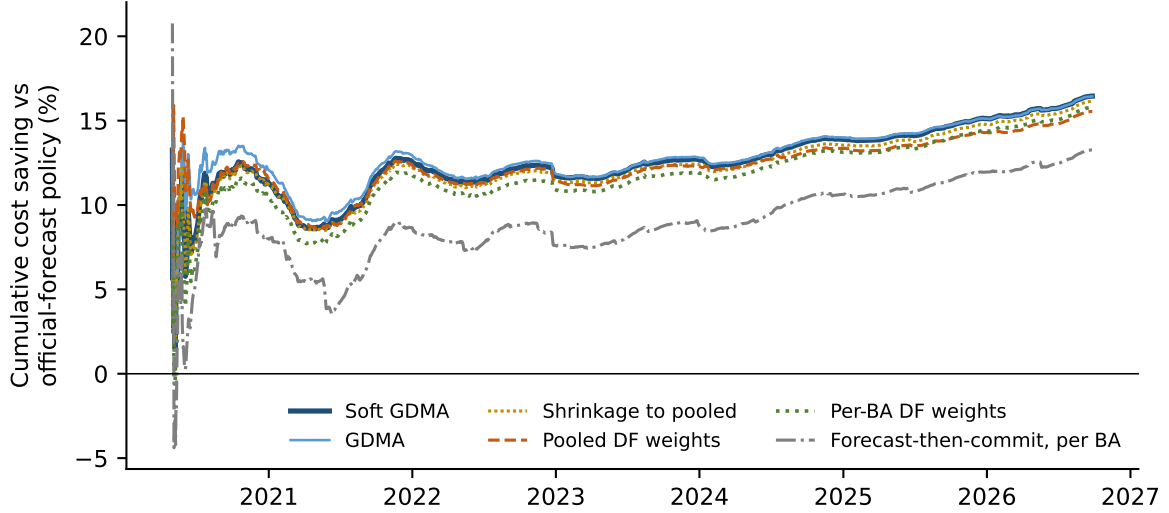


Figure 3: Cumulative system-wide cost saving relative to the official-forecast policy ($W = 28$). Higher is better.

6.7. Extreme events and robustness

In Table 5, during Winter Storm Uri and the 2023 summer heat, the decision-focused combinations cost 4–7% and 16–19% less than the official-forecast policy. For Uri, ERCOT, and on two days SPP and MISO, shed load (FERC, NERC and Regional Entity Staff, 2021) and EIA-930 reports served demand, so anticipating true demand is scored as over-commitment; Section 7 repeats the analysis without the load-shed BA-days. During Winter Storm Elliott the operators’ forecasts beat every candidate rule over the six days and every combination is 17–30% more expensive than the official policy, as expected for an abrupt cold snap with large forced outages (FERC, NERC and Regional Entity Staff, 2023) that weights learned from earlier weeks cannot anticipate; Section 7 tests whether weather-driven candidates narrow this gap.

Table 6 replaces the calibrated ratios by a common τ , from the symmetric $\tau = 0.5$ to the highly asymmetric $\tau = 0.95$. Savings change little (soft GDMA saves 17–19%) and neither does the ranking: two-step clustering is cheapest in every column, soft GDMA and GDMA follow within 0.5%, then shrinkage, per-BA and pooled weights. The advantage over the better forecast-then-commit method grows with the asymmetry, from 1.4% at $\tau = 0.5$ to 2.8% at $\tau = 0.95$, as the best combination for the decision departs further from the most accurate one.

7. Capacity commitment on five continents

7.1. A global panel of grid areas

We assemble public hourly demand data for 125 grid areas on five continents (Table 7). *North America* is the 46 U.S. balancing authorities of Section 6 (EIA Form 930, 2019–2026), the only systems whose operators’ day-ahead forecasts are in the data; elsewhere the reference rule is a model forecast (Section 7.2). *Europe* contributes 33 national systems from Energy-Charts (Fraunhofer Institute for Solar Energy Systems ISE, 2026), including cross-border flows, and *Oceania* the five regions of Australia’s National Electricity Market (Australian Energy Market Operator, 2026), both 2018–2026. *Asia* contributes 31 Chinese provinces for 2018 (Wu and Kan, 2023), four Taiwanese grid areas for 2018 to mid-2022

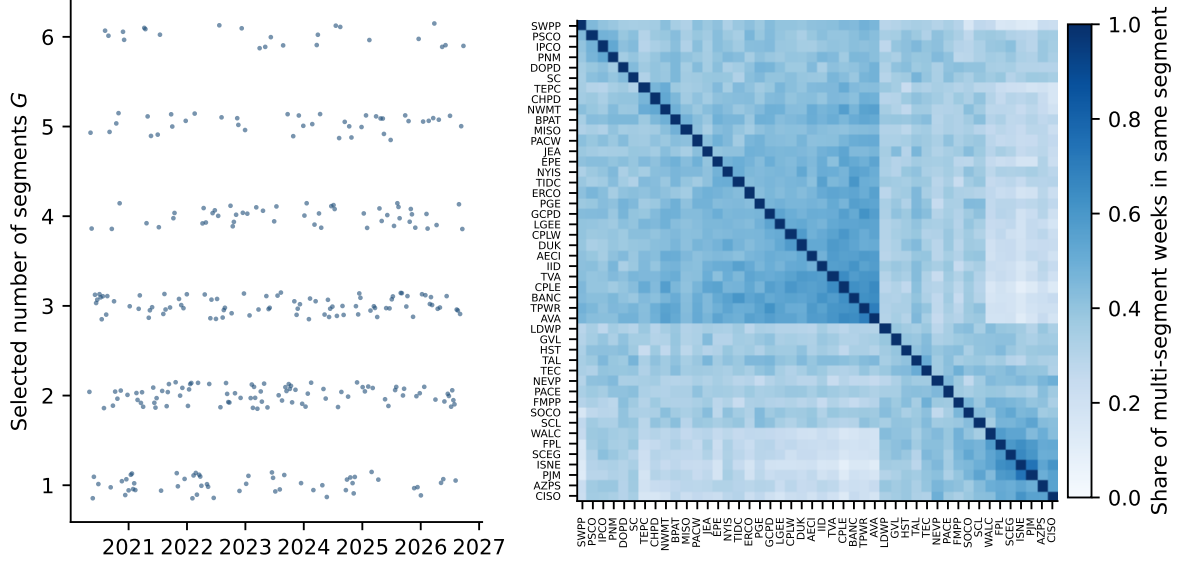


Figure 4: Left: number of segments selected by GDMA in each weekly re-estimation ($W = 28$). Right: among weeks with $G \geq 2$, share of weeks in which two BAs share a segment.

(Fu and Chen, 2023) and five Thai regions for 2023–2024 (Bunnak, 2025); *Africa* is Algeria (Sonelgaz), 2018 to January 2020 (Rania, 2023). All series are converted to hourly UTC, cleaned as in Section 6, and placed on a common calendar from 1 January 2018 to 26 September 2026. The average demands sum to 1.66 TW, which is not demand served at any one time because areas are observed in different periods. The panel is deliberately unbalanced: several Asian and African systems have only one or two years of data, as is typical of emerging economies, and an area contributes only on days with data. Other large systems (South Africa, India, Japan, South Korea) could not be retrieved (Section 9).

Soft GDMA has a lower cost than the reference in every European area (0.84–0.95); the largest gains are in three western U.S. balancing authorities (Tucson, the California ISO and the Lower Colorado area, 0.45–0.59), and 25 of the 125 areas, 13 of them in China, have a higher cost (Figure 5).

7.2. Candidates, costs and competitors

Every area receives eleven candidate forecasts: the seven of Section 6; a zero-shot forecast from the pretrained foundation model Chronos-Bolt (small; Ansari et al., 2024), which takes the 28 days of demand known at decision time, forecasts 48 hours and contributes the last 24; and three weather-aware global LightGBM models. Chronos was pretrained on public series including electricity load, so some of our data may be in its corpus. Outside the United States candidate 0, the *reference forecaster*, is the global LightGBM forecast, a strong rule distinct from the best single candidate. The temperature model adds to the LightGBM features the operating day’s hourly, mean, minimum and maximum 2-m temperature, heating and cooling degrees (base 18 °C) and the previous day’s mean, from NASA POWER (NASA Langley Research Center, 2026) averaged over one to five load centres per area (169 locations). Realised temperature would be an *ex post* forecast in the terminology of Hong and Fan (2016), so the weather models use *archived day-ahead forecasts* of NOAA’s Global Forecast System (NOAA National Centers for Environmental Prediction, 2026b): for day d , the 00 UTC run of day $d - 1$ at lead times of 24 to 48 hours, interpolated to the load centres and to hourly resolution (the previous

Table 5: System cost during extreme events relative to the official-forecast policy ($W = 28$): Winter Storm Uri (10–20 February 2021), Winter Storm Elliott (21–26 December 2022) and the summer heat of 15 July–15 August 2023. Lower is better: values below 1 mean a lower cost than the benchmark.

Method	Uri	Elliott	Heat wave 2023
Official forecast	1.000	1.000	1.000
Equal weights	1.292	2.984	1.501
Best single (per BA)	0.982	1.204	0.828
Pooled DF weights	0.955	1.303	0.816
Per-BA DF weights	0.941	1.189	0.836
Shrinkage to pooled	0.932	1.189	0.834
Two-step k -means	0.931	1.180	0.811
Forecast-then-commit, per BA	1.006	1.512	0.829
Forecast-then-commit, grouped	0.974	1.369	0.807
GDMA	0.951	1.275	0.818
Quantile-regression comb.	1.011	1.488	0.821
GDMA, minimum hold-out rule	0.935	1.169	0.825
Soft GDMA	0.931	1.205	0.827

day’s temperature is the forecast for day $d - 1$ issued on day $d - 2$). The run is available at about 04–05 UTC, before a 10:00-local gate closure in the Americas, Europe and Algeria but after it in East Asia and Australia, so only the first group uses weather forecasts. The archive starts in January 2021; before then the weather models use their other features. The GFS forecasts have a root-mean-squared error of 2.3°C against realised temperatures (1.9 in Europe, 2.2 in North America, 2.7 in Asia); a version with realised temperatures is an upper bound. The *multi-variable model* adds, from the same GFS runs, forecast 2-m relative humidity, 10-m wind speed and downward short-wave radiation, and from the GEFSv12 ensemble (Zhou et al., 2022; NOAA National Centers for Environmental Prediction, 2026a) the standard deviation of 2-m temperature at 0.5 degrees (from October 2020). The third weather candidate uses this spread for the commitment margin: it multiplies the multi-variable model’s quantile margin by the ratio of the day’s spread to its mean over the previous 28 days, clipped to $[0.5, 3]$ (*ensemble margin*). The rule was motivated by Winter Storm Elliott and uses a raw, uncalibrated spread from a single archive, so it is exploratory; its constants were set once, untuned. The spread says little about the error on an average day (rank correlation 0.04 across area-days, 0.08 within areas) but is large in unusual weather. Mean absolute errors, scaled by mean demand, are 3.5% for the temperature and multi-variable models (also with exact temperatures), 3.6% for LightGBM, 3.8% for Chronos, 4.5% for the official/LightGBM candidate and 6–8% for the naive rules.

Critical ratios follow the flexibility rule of Section 6 wherever interchange is measured (the United States, 31 European countries, and Taiwan, an island grid with zero flexibility); the rest receive the midpoint $\tau = 0.875$. A common τ gives the same ranking (Section 6.7, Table D.22).

Beyond the benchmarks of Section 6, we add a *decision-focused neural gate*, a two-hidden-layer perceptron (NGPARAMS parameters with eleven candidates). From features at the weekly origin (each candidate’s decision cost and forecast error over the last 7 and 28 known days, the critical ratio, recent demand, data share, continent and season)

Table 6: Sensitivity to the cost asymmetry ($W = 28$): system cost relative to the official-forecast policy when every BA has the same critical ratio τ , and under the calibrated heterogeneous ratios. Lower is better: values below 1 mean a lower cost than the benchmark.

Method	$\tau = 0.5$	$\tau = 0.8$	$\tau = 0.9$	$\tau = 0.95$	Calibrated
Official forecast	1.0000	1.0000	1.0000	1.0000	1.0000
Equal weights	1.5702	1.5975	1.6418	1.6623	1.6902
Best single (per BA)	0.8459	0.8240	0.8346	0.8562	0.8585
Pooled DF weights	0.8351	0.8145	0.8246	0.8424	0.8446
Per-BA DF weights	0.8349	0.8134	0.8226	0.8389	0.8422
Shrinkage to pooled	0.8307	0.8100	0.8191	0.8345	0.8385
Two-step k -means	0.8268	0.8055	0.8142	0.8294	0.8348
Forecast-then-commit, per BA	0.8470	0.8266	0.8420	0.8687	0.8673
Forecast-then-commit, grouped	0.8423	0.8213	0.8335	0.8561	0.8548
GDMA	0.8297	0.8092	0.8171	0.8337	0.8356
Quantile-regression comb.	0.8457	0.8268	0.8495	0.8904	0.8834
GDMA, minimum hold-out rule	0.8317	0.8096	0.8187	0.8362	0.8383
Soft GDMA	0.8303	0.8088	0.8171	0.8326	0.8355

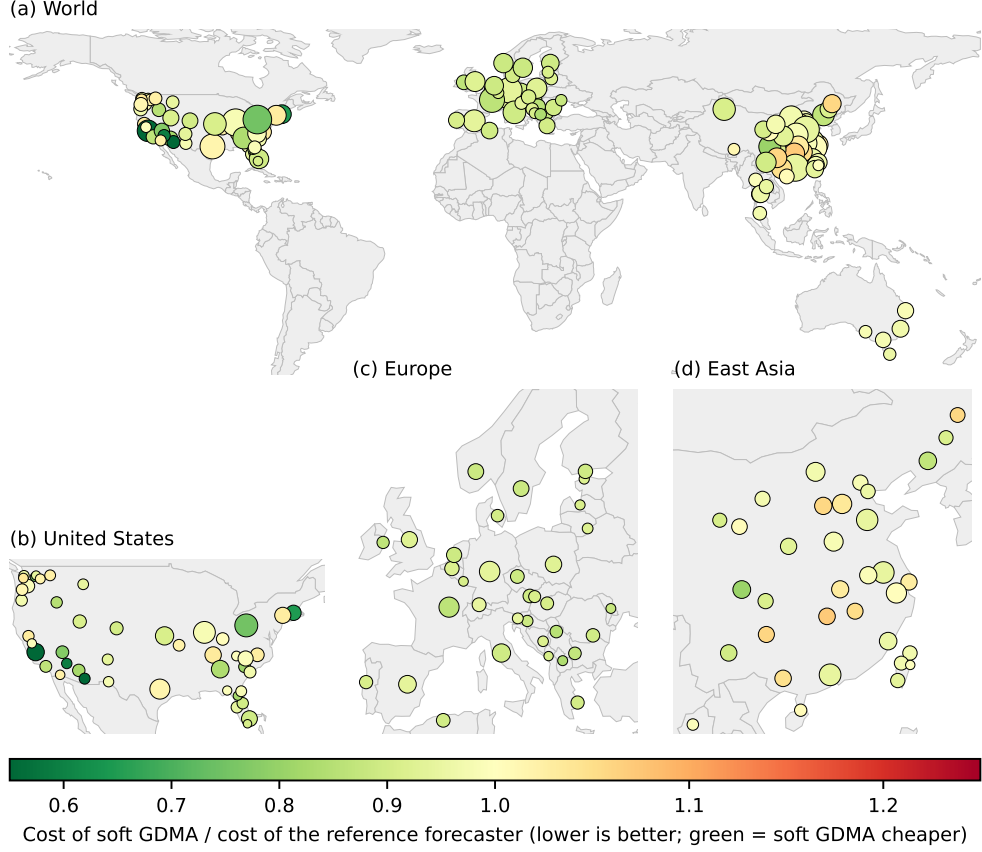
Table 7: The global panel.

Continent	Systems	Areas	Period	Mean demand
North America	U.S. balancing authorities (EIA-930)	46	2019–2026	455 GW
Europe	National systems (Energy-Charts)	33	2018–2026	339 GW
Oceania	NEM regions, Australia (AEMO)	5	2018–2026	21 GW
Asia	China (31 provinces)	31	2018	
	Taiwan (4 grid areas)	4	2018–2022	839 GW
	Thailand (5 regions)	5	2023–2024	
Africa	Algeria (Sonelgaz)	1	2018–2020	7 GW
Total		125	2018–2026	1.66 TW

it produces combination weights through a softmax, as in mixture-of-experts averaging (Gong et al., 2026). It is trained end to end on the same smoothed newsvendor cost, on up to 52 recent weeks with fully observed outcomes, and re-trained every four weeks; the four most recent weeks serve for early stopping, and two initialisations are averaged. It is untuned. Unlike GDMA it lets weights vary continuously with conditions; like GDMA it uses only information available at decision time.

7.3. Results

Table 8 shows that the main U.S. findings carry over to the world panel, with smaller savings. Worldwide, soft GDMA lowers the commitment cost by 11.6% relative to the reference forecaster and by 4.2 percentage points relative to the best single candidate; its fair variant lowers it by 11.9%. The 90% model confidence set (Hansen et al., 2011) contains four methods (two-step clustering, GDMA, soft GDMA and fair soft GDMA); it excludes shrinkage towards pooled weights ($p = 0.098$) and per-area weights only narrowly, and the neural gate, the quantile-regression combination and the best single candidate more clearly. Within the set the differences are below 0.4%; fair soft GDMA is the cheapest in total (0.881), significantly so against soft GDMA. By continent a GDMA



Marker area proportional to the square root of mean demand. Marker position: mean of the load centres used for the area's weather.

Figure 5: Where soft GDMA saves cost. Each marker is one of the 125 grid areas, coloured by the cost of soft GDMA relative to the reference forecaster ($W = 28$; lower is better, values below 1 mean a lower cost). Marker size shows mean demand.

variant is cheapest in North America and Africa, the neural gate nominally in Europe and Oceania (0.896 and 0.964, against 0.897 and 0.972 for the best GDMA variant), and pooled weights in Asia (0.929 against 0.963–0.965), where most areas have little data.

Three comparisons stand out. First, decision-focused combination on the simplex dominates forecast-then-commit and the unconstrained quantile-regression combination: combining forecasts for accuracy and adding a quantile margin saves only 1–3% worldwide, and the quantile-regression combination costs more than the reference (1.016) because, with eleven collinear candidates and few weeks of data, freeing the weights from the simplex costs more in variance than it gains; both are poor in Asia. Second, the neural gate is clearly worse than the simple decision-focused rules (0.956 against 0.884 worldwide), although competitive in Europe and Oceania; flexibility alone does not solve the problem, but an untuned gate says little about the best neural design. Third, better candidates and better combination are complements. On their own the temperature model costs 0.965 of the reference, the multi-variable model 0.960, its ensemble-margin version 0.981, LightGBM 0.988 and the zero-shot foundation model 1.434, which must forecast two days ahead. Adding the temperature model lowers the cost of soft GDMA from 0.906 to 0.892 and the two multi-variable candidates to 0.884 (Table 9); with realised temperatures it is 0.876. The method ranking is the same for every candidate set and similar across windows and critical ratios (Tables D.21 and D.22).

Table 8: Out-of-sample commitment cost on five continents ($W = 28$ days, May 2018 to September 2026, eleven candidates including the three weather models), relative to the reference forecaster. Bold: lowest in the column. Stars: Diebold–Mariano test of equal daily cost against soft GDMA (Newey–West, 7 lags); *, **, ***: $p < 0.10, 0.05, 0.01$. †: member of the 90% model confidence set. Lower is better: values below 1 mean a lower cost than the benchmark.

Method	World	North America	Europe	Oceania	Asia	Africa
Reference forecaster ^a	1.000***	1.000***	1.000***	1.000***	1.000	1.000***
Equal weights	1.217***	1.437***	1.062***	0.993***	1.033***	0.977***
Best single (per area)	0.923***	0.872***	0.946***	1.007***	1.018***	0.904
Pooled DF weights	0.902***	0.874***	0.915**	0.982**	0.929***	0.916
Per-area DF weights	0.895***	0.848***	0.913***	0.988***	0.985***	0.920**
Shrinkage to pooled	0.888***	0.844***	0.906***	0.983***	0.963	0.910
Two-step k -means	0.884†	0.849***	0.897	0.975	0.941***	0.888
Quantile-regression comb.	1.016***	1.005	0.989***	0.986**	1.222**	1.030***
Forecast-then-commit, per area	0.988**	0.985	0.934***	0.987***	1.286**	0.929
Forecast-then-commit, grouped	0.966*	0.978	0.899	0.978	1.249*	0.914
Neural gate (deep MoE)	0.956***	1.007***	0.896	0.964***	1.008	0.899
GDMA	0.884†	0.843***	0.901	0.972**	0.951***	0.879*
Soft GDMA	0.884†	0.839	0.902	0.975	0.965	0.895
Fair soft GDMA	0.881***†	0.837**	0.897***	0.974	0.963	0.892
Grid areas	125	46	33	5	40	1

^a The operator’s official forecast where one is published (United States), otherwise the global LightGBM forecast.

Extreme events. In the U.S. analysis (Table 5) the operators’ own forecasts beat every combination during Winter Storm Elliott, the abrupt cold snap of 21–26 December 2022 (FERC, NERC and Regional Entity Staff, 2023). Over the six days soft GDMA has a 20.0% higher cost than the official forecasts without weather candidates, 14.7% with GFS temperature forecasts and 9.5% with the multi-variable candidates, and 11.1% with realised temperatures (Table 10); six days are too few to rank candidate sets. Excluding the BA-days on which TVA and Duke shed load (FERC, NERC and Regional Entity Staff, 2023), which understate demand, the excess is 8.4%. Only the multi-variable model with the ensemble margin beats the operators, at a 7% lower cost (0.929), because the GEFS spread rose on 23 December, the costliest day, and widened the margin by a median factor of 1.4. This is exploratory: the rule was designed after the gap was known and is evaluated on the event that motivated it; it is the most expensive weather candidate over the whole period (0.981), and the combination rules do not give it enough weight. During Winter Storm Uri, soft GDMA has an 11–13% lower cost than the operators in every candidate set, also excluding ERCOT’s, SPP’s and MISO’s load-shed days.

As a single storm is a weak basis, we also examine all extreme-temperature area-days, fixed beforehand as those with daily mean temperature below its 2.5% or above its 97.5% quantile (Table 11). On cold days soft GDMA has a 12% lower cost than the reference worldwide (90% interval 0.83–0.93, block bootstrap over days) and on hot days 19% lower (0.79–0.84); in the United States 10% and 25% lower. Elliott is thus atypical: an abrupt cold snap with large forced outages, which weights cannot anticipate.

Tail risk. To see the tail, Table 12 reports the average of the costliest 5% of days (CVaR_{95}) and the costliest week of the world’s daily cost. Soft GDMA’s costliest week costs 16% less than the reference’s (0.840) and its CVaR_{95} is 9.5% lower (0.905), so its saving is not bought with a heavier tail. Forecast-then-commit and quantile-regression rules fail in the

Table 9: World commitment cost relative to the reference forecaster for four candidate sets ($W = 28$): eight candidates without weather; nine with the temperature model fed archived GFS forecasts; eleven with, in addition, the multi-variable model and its ensemble-margin version (main specification); and nine with realised temperatures (upper bound). Lower is better: values below 1 mean a lower cost than the benchmark.

Method	No weather	GFS temperature	GFS/GEFS multi-variable (main)	Realised temperature
Reference forecaster ^a	1.000	1.000	1.000	1.000
Equal weights	1.380	1.312	1.217	1.308
Best single (per area)	0.942	0.929	0.923	0.915
Pooled DF weights	0.929	0.908	0.902	0.894
Per-area DF weights	0.918	0.903	0.895	0.886
Shrinkage to pooled	0.911	0.896	0.888	0.879
Two-step k -means	0.904	0.889	0.884	0.875
Quantile-regression comb.	1.038	1.022	1.016	1.005
Forecast-then-commit, per area	1.011	0.994	0.988	0.977
Forecast-then-commit, grouped	0.988	0.971	0.966	0.957
Neural gate (deep MoE)	0.968	0.958	0.956	—
GDMA	0.905	0.889	0.884	0.875
Soft GDMA	0.906	0.892	0.884	0.876
Fair soft GDMA	0.905	0.887	0.881	0.871

tail: their CVaR₉₅ is 55–63% higher and their costliest week nine times as costly, mostly in the first week of January 2019, when the 46 U.S. balancing authorities enter with a few days of data (more than twice as costly even without each area’s first eight weeks). With a scarcity-level critical ratio of 0.99 for every area, soft GDMA’s total cost is 11% lower (0.888), its CVaR₉₅ 8% lower and its costliest week 10% lower; forecast-then-commit and the quantile-regression combination cost more than the reference in total (1.18–1.53) and 14–15 times as much in their costliest week. Equal weights have the lightest tail at this ratio (costliest week 0.66) but a higher total cost (1.071).

New and data-poor systems. Fifty-two areas enter after the evaluation starts (the 46 U.S. balancing authorities in January 2019, the five Thai regions in January 2023, one European system) and have only a few weeks of history in their first eight weeks (Table 13). No combination does well then: all except the quantile-regression combination (0.976) cost more than the reference. Pooled weights cost 61% more, forecast-then-commit 39–40% more, and the neural gate, whose features are not yet informative, 2.6 times as much. Plain GDMA (1.053) and fair soft GDMA (1.058) are closest to the reference, followed by soft GDMA and the best single candidate (1.069, each); per-area weights and shrinkage cost 11% more, and two-step clustering (1.170) more still because it clusters weights estimated from a handful of days. The quantile-regression combination beats the reference for entering areas but not for established ones (1.016); we cannot explain this and the entry sample is small. Where an official forecast exists, an operator should keep committing on it for a new unit’s first weeks.

Segments across continents. GDMA uses more than one segment in 89% of weekly re-estimations (two in 20%, three in 22%, four in 21%), much more than in the U.S. panel; soft GDMA shrinks towards the segment rule with an average $\kappa = 0.39$. Segments are not continents: among weeks with several segments, two systems on the same continent share

Table 10: Winter Storm Elliott (21–26 December 2022, 46 U.S. balancing authorities): commitment cost relative to the operators’ official day-ahead forecasts, by candidate set ($W = 28$), and in the main set without the BA-days on which load was shed. Lower is better: values below 1 mean a lower cost than the benchmark.

	No weather	GFS temperature	GFS/GEFS multi-variable	Realised temperature	Main set, no shed days
<i>Combination rules</i>					
Soft GDMA	1.200	1.147	1.095	1.111	1.084
Fair soft GDMA	1.176	1.084	1.112	1.091	1.103
GDMA	1.250	1.216	1.071	1.121	1.056
Two-step k -means	1.197	1.143	1.128	1.105	1.123
Shrinkage to pooled	1.198	1.136	1.138	1.121	1.128
Per-area DF weights	1.169	1.127	1.138	1.116	1.128
Pooled DF weights	1.299	1.177	1.169	1.159	1.143
Best single (per area)	1.239	1.144	1.136	1.131	1.100
Quantile-regression comb.	1.361	1.354	1.367	1.345	1.351
Forecast-then-commit, grouped	1.395	1.170	1.171	1.198	1.152
<i>Weather candidates on their own</i>					
Temperature model	–	1.100	1.100	1.112	–
Multi-variable model	–	–	1.074	–	–
Multi-variable, ensemble margin	–	–	0.929	–	–

one only 37–49% of the time, European and African systems 52%, and North American areas with European or Asian ones 19–21%. The segmentation fits differences in which forecasting systems work, notably whether an operator’s own forecast is available and informative, which a planner could not specify in advance.

7.4. Who benefits? A fairness audit

Table 14 asks how the gains are distributed, relative to the reference forecaster and to each area’s own best single candidate. Against the reference the picture is mixed: even the best rules make 17–21% of areas worse off, because the reference is often strong, and rules that impose a common rule or learn a flexible global map do worse (neural gate 34%, forecast-then-commit 41–45%, quantile-regression combination 58%). Against each area’s own best candidate the groups separate: soft GDMA makes only 6% of areas worse off and its fair variant 3%, pooled weights 24%, per-area weights and two-step clustering 14%, the neural gate 42%, forecast-then-commit 52–60% and the quantile-regression combination 86%. The median area in the worst-served continent is 2.2–2.8% better off under the GDMA variants and clustering rules, but worse off under the neural gate (1.010), forecast-then-commit (1.047–1.053) and the quantile-regression combination (1.127). Data-rich and data-poor systems differ: areas with at least two years of data save 12% under soft GDMA, those with less (the Chinese provinces and Algeria) only 4%, and per-area weights save almost nothing for the latter (0.983); only pooled weights do nearly as well for both (0.922 against 0.901), at a higher cost for the rich ones.

Fair soft GDMA, which counts every area’s relative improvement equally, is also the cheapest in total (0.881 against 0.884, significant at the 1% level but negligible and within the confidence set). Fairness thus costs essentially nothing in total and shifts the distribution: the share of areas worse off than their own best candidate halves (6% to 3%), and the 10th and 90th percentiles of the ratio become 0.840 and 1.022. Weighting by relative cost keeps the largest systems from dictating the segment rules, consistent with

Table 11: Commitment cost relative to the reference forecaster on extreme-temperature area-days ($W = 28$): days on which an area’s daily mean temperature is below its 2.5% or above its 97.5% quantile. Intervals: moving-block bootstrap over days (7-day blocks, 1,000 replications). Lower is better: values below 1 mean a lower cost than the benchmark.

	World cold	World hot	U.S. cold	U.S. hot
Reference forecaster ^a	1.000	1.000	1.000	1.000
Equal weights	1.239	0.975	1.620	1.066
Best single (per area)	0.952	0.879	0.962	0.810
Pooled DF weights	0.903	0.827	0.975	0.776
Per-area DF weights	0.901	0.822	0.923	0.754
Shrinkage to pooled	0.889	0.817	0.912	0.751
Two-step k -means	0.876	0.827	0.913	0.768
Quantile-regression comb.	1.138	0.971	1.144	0.864
Forecast-then-commit, per area	1.048	0.910	1.106	0.820
Forecast-then-commit, grouped	0.954	0.885	1.012	0.808
GDMA	0.877	0.818	0.905	0.748
Soft GDMA	0.879	0.814	0.898	0.747
Fair soft GDMA	0.884	0.815	0.904	0.752
Soft GDMA / reference, 90% CI	[0.83, 0.93]	[0.79, 0.84]	[0.86, 0.94]	[0.71, 0.79]
Soft GDMA / best single, 90% CI	[0.90, 0.94]	[0.91, 0.94]	[0.91, 0.96]	[0.89, 0.95]
Area-days	9,624	9,623	3,542	3,540

^a As in Table 8.

Section 4, under which fair and efficient criteria share their minimiser; these data cannot tell whether segments are exact.

Income. Each area is assigned its country’s GDP per capita at purchasing-power parity and World Bank income group (World Bank, 2026) (Taiwan: IMF World Economic Outlook (International Monetary Fund, 2026)); 83 areas are in high-income and 42 in upper-middle-income countries. For U.S. balancing authorities we also use the per-capita personal income of the states served (U.S. Bureau of Economic Analysis, 2026), weighted by approximate service-territory shares (Table 15).

Areas in upper-middle-income countries save less than high-income ones under every decision-focused simplex method (soft GDMA 4.9% against 12.2%), but this largely coincides with the data gap, since the group is dominated by Chinese provinces with a single year of data. The rank correlation between cost ratio and GDP per capita is -0.12 for soft GDMA (90% bootstrap interval -0.26 to 0.03) and -0.11 for its fair variant, so richer areas gain slightly more but the interval includes zero; it is positive and significant for pooled weights (0.19 , interval 0.04 to 0.34), which favour poorer areas, and not different from zero for the neural gate. Among the 88 areas with at least two years of data it is positive for every method checked (0.17 for soft GDMA, 0.18 for its fair variant, 0.23 for pooled weights, 0.39 for the neural gate): richer areas gain *less*. The unstable sign suggests the correlation reflects data length and continent, with which income is confounded. In the United States, balancing authorities in the highest third of state income gain more in cost-weighted terms (0.68 under soft GDMA against 0.93 in the lowest third), reflecting a few large systems with weak official forecasts; the rank correlation with state income is 0.03 (90% interval -0.24 to 0.31). We find no evidence that GDMA systematically favours richer or poorer systems, but with 125 areas, income measured at country and state level, and strong confounding with data length, moderate gradients cannot be ruled

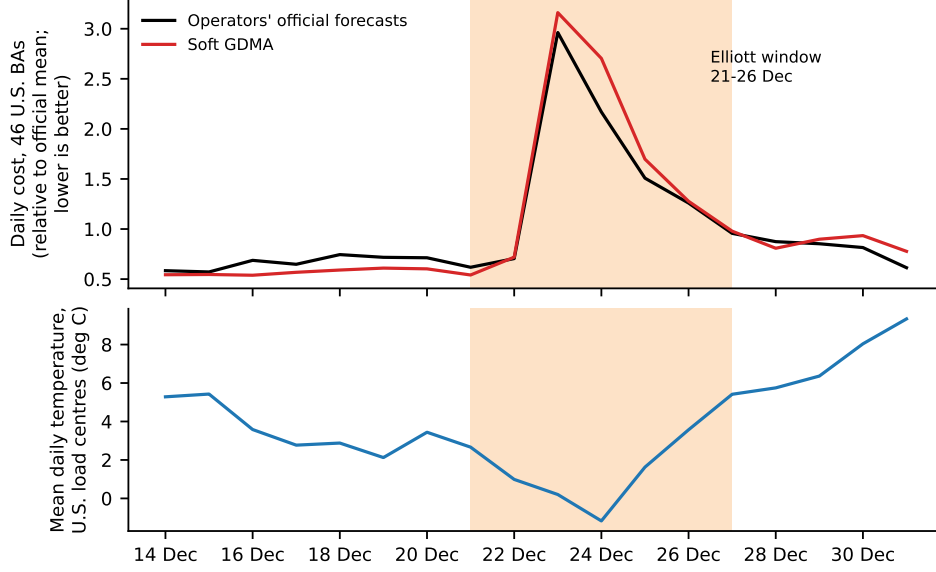


Figure 6: Winter Storm Elliott, 14–31 December 2022: daily commitment cost of the 46 U.S. balancing authorities (relative to the mean daily cost under the operators’ forecasts; lower is better) and mean temperature of their load centres. Shaded: the FERC–NERC window.

out.

8. Where to install batteries

We now solve the planning problem (4) on the global panel. For each commitment rule we take its out-of-sample hourly shortfalls and surpluses ($W = 28$, eleven candidates with GFS and GEFS weather forecasts, day-ahead gate-closure information), place a four-hour battery with 85% round-trip efficiency in every grid area, and compute the annual value of battery power on a grid from 0 to 15% of the area’s mean demand. Costs are converted to money by valuing one MWh of idle committed capacity at an illustrative \$10, so a shortfall costs between \$40 and \$190 per MWh depending on the area’s critical ratio; all dollar figures scale proportionally. The battery is dispatched greedily against the given commitment path, charging from idle committed capacity and discharging into shortfalls; the commitment is not re-optimised with the battery in place, which would lower its value. The exercise values a single stream, cover for day-ahead commitment errors, and ignores arbitrage, adequacy, renewable firming and network deferral, which dominate real storage decisions. It illustrates how the operational rule changes the value and location of storage and is not a siting recommendation; annual values average over each area’s own evaluation period.

8.1. Batteries do not pay for themselves on forecast errors alone

The first increment of battery power is worth \$31,000–\$45,000 per MW-year in the median area of each continent (depending on the rule), and at most \$73,000 (in the PJM interconnection). Annualised costs of four-hour batteries are well above this range (we use \$100,000–\$200,000 per MW-year), so the cost-optimal battery size for covering commitment errors alone is zero in every area and for every rule. Storage is deployed for other reasons; covering commitment errors is a by-product. The planning question is

Table 12: Tail risk of the world’s daily commitment cost relative to the reference forecaster ($W = 28$): total, average of the costliest 5% of days (CVaR₉₅) and costliest week, for the calibrated critical ratios, without each area’s first eight weeks, and with a scarcity-level critical ratio of 0.99 for every area. Lower is better: values below 1 mean a lower cost than the benchmark.

Method	Calibrated τ_i (main)			Without first 8 weeks of each area		$\tau_i = 0.99$ for every area		
	CVaR ₉₅		Worst	CVaR ₉₅	Worst	CVaR ₉₅		Worst
	Total	daily	week	daily	week	Total	daily	week
Reference forecaster ^a	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Equal weights	1.217	1.136	1.068	1.114	1.019	1.071	0.772	0.660
Best single (per area)	0.923	0.981	0.946	0.971	0.905	1.043	1.312	1.135
Pooled DF weights	0.902	0.914	0.919	0.901	0.830	0.886	0.839	0.783
Per-area DF weights	0.895	0.920	0.829	0.908	0.845	0.909	0.946	0.904
Shrinkage to pooled	0.888	0.902	0.845	0.893	0.805	0.893	0.920	0.904
Two-step k -means	0.884	0.892	0.853	0.884	0.785	0.879	0.858	0.786
Quantile-regression comb.	1.016	1.634	8.955	1.152	2.316	1.531	3.092	14.704
Forecast-then-commit, per area	0.988	1.603	8.968	1.117	2.313	1.247	2.435	14.564
Forecast-then-commit, grouped	0.966	1.554	8.995	1.067	2.274	1.180	2.279	14.521
Neural gate (deep MoE)	0.956	0.943	0.996	0.859	0.794	—	—	—
GDMA	0.884	0.902	0.866	0.893	0.804	0.888	0.912	0.760
Soft GDMA	0.884	0.905	0.840	0.895	0.820	0.888	0.922	0.904
Fair soft GDMA	0.881	0.898	0.848	0.888	0.826	0.882	0.893	0.904

therefore not whether but *where* to place the storage that energy-transition programmes already mandate, and how that depends on how operators commit.

Figure 7 shows the value of the first MW in each area: \$19,000 to \$54,000 per MW-year, with continental medians from \$31,000 (Asia) to \$41,000 (Oceania), all well below the battery costs we consider.

8.2. Better commitment rules reduce the value of storage

Without storage (Figure 8, Table 16), soft GDMA commits the five continents’ capacity for \$11.3 billion a year under the stylised cost model, against \$12.3 billion for the reference forecaster, a difference of \$1.0 billion or 8%. Adding batteries lowers every curve: the reference rule needs 34 GW of optimally sited batteries to reach the cost that soft GDMA achieves without any, forecast-then-commit needs 17 GW and per-area decision-focused weights 5 GW, while pooled weights and plain GDMA are as cheap as soft GDMA without storage in this metric. The battery-equivalent is not the value of the better rule: 34 GW would cost far more than the \$1.0 billion difference in commitment cost. The point is the interaction: the better the rule, the less is left for storage to absorb, and a planner who values storage by today’s errors overstates it.

8.3. The siting decision depends on the commitment rule

Table 16 also shows how a 10 GW programme should be split. Under soft GDMA, 4.3 GW go to North America, 2.6 GW to Europe, 0.7 GW to Oceania and 2.5 GW to Asia, and Africa receives almost nothing. Using the reference forecaster’s error profile instead, North America receives 5.1 GW, about 19% more, because the U.S. balancing authorities’ own forecasts leave the largest shortfalls; forecast-then-commit would send 5.0 GW there.

Table 17 shows that siting on the wrong profile costs little: when operations follow soft GDMA, a 1 GW programme sited with the reference rule’s profile delivers \$43 million a year instead of \$45 million, a 4% loss; the loss is 4% at 5 GW, 3% at 10 GW and 2% at 20 GW. Location therefore matters less than having a good commitment rule, but it is

Table 13: Commitment cost of areas entering the panel, relative to the reference forecaster ($W = 28$, eleven candidates). Lower is better: values below 1 mean a lower cost than the benchmark.

Method	First 8 weeks after entry ^b	All other weeks
Reference forecaster ^a	1.000	1.000
Equal weights	2.477	1.206
Best single (per area)	1.069	0.922
Pooled DF weights	1.614	0.896
Per-area DF weights	1.111	0.893
Shrinkage to pooled	1.115	0.886
Two-step k -means	1.170	0.881
Quantile-regression comb.	0.976	1.016
Forecast-then-commit, per area	1.394	0.985
Forecast-then-commit, grouped	1.400	0.962
Neural gate (deep MoE)	2.599	0.942
GDMA	1.053	0.882
Soft GDMA	1.069	0.882
Fair soft GDMA	1.058	0.879
Grid areas entering	52	

^b The first eight weeks after an area has at least one day of data in the estimation window.

not irrelevant: the difference is a few per cent of the programme’s value, concentrated in North America.

Siting on estimated value curves also carries estimation risk. Table 18 sites batteries on the first half of the evaluation period and values them on the second: soft GDMA’s own profile captures about three quarters of the infeasible oracle’s value at 10 GW (\$247 million of \$336 million), and the reference rule’s profile captures 6% less than that at 10 GW and 8% less at 20 GW. The two sources of loss, the choice of rule and the estimation of the profile, are of similar size for programmes of 5–10 GW.

8.4. Fair siting

Efficient siting concentrates batteries where the monetary value is highest, so in small programmes some continents receive nothing: with 1 GW, the worst-served continent’s relative benefit is zero (Table 19). Floors proportional to each continent’s share of demand, the “fair share” rule, raise it only slightly (from 1.4% to 1.7% of its commitment cost with 10 GW) and give up 3–7% of the programme’s value. Maximin siting, which directs every increment to the continent that has gained least so far, raises it much more, to 2.8% with 10 GW and from zero to 0.3% with 1 GW, at a price of fairness of 7–9%. Fairness in storage siting is therefore not free here: protecting the worst-served continent costs between a thirtieth and a tenth of the programme’s value, and only for one value stream and with the continent as the unit of fairness.

9. Managerial implications and conclusion

Implications for system operators and planners. Five lessons emerge for organisations that coordinate many operating units. First, learn the *decision rule*, not the forecast: combining for accuracy and adding a margin leaves 2–3 points of the saving on the table in the United States and 8–10 points worldwide. Second, planners need not choose between one rule for all systems and a bespoke rule for each: a handful of segments, or even one

Table 14: Distribution of the benefits across the 125 grid areas ($W = 28$, eleven candidates): percentiles of each area’s cost relative to the reference forecaster, share of areas made worse off than under the reference and under their own best single candidate, median ratio in the worst-served continent, and cost ratios for data-rich and data-poor areas. Cost ratios and shares: lower is better.

Method	Per-area cost ratio			Areas harmed vs		Worst	Data-rich	Data-poor
	P10	Median	P90	ref.	best single	continent ^d	areas	areas ^e
Reference forecaster ^a	1.000	1.000	1.000	0.00	0.62	1.000	1.000	1.000
Equal weights	0.963	1.050	1.652	0.71	0.79	1.337	1.231	1.024
Best single (per area)	0.876	0.976	1.047	0.38	–	1.010	0.917	1.006
Pooled DF weights	0.856	0.935	1.037	0.19	0.24	0.984	0.901	0.922
Per-area DF weights	0.852	0.952	1.034	0.27	0.14	0.991	0.888	0.983
Shrinkage to pooled	0.848	0.941	1.027	0.21	0.09	0.987	0.883	0.958
Two-step k -means	0.846	0.927	1.020	0.18	0.14	0.974	0.880	0.936
Quantile-regression comb.	0.917	1.032	1.221	0.58	0.86	1.127	1.010	1.097
Forecast-then-commit, per area	0.887	0.985	1.232	0.45	0.60	1.047	0.982	1.064
Forecast-then-commit, grouped	0.868	0.972	1.214	0.41	0.52	1.053	0.962	1.028
Neural gate (deep MoE)	0.872	0.956	1.109	0.34	0.42	1.010	0.952	1.008
GDMA	0.854	0.934	1.015	0.17	0.11	0.972	0.879	0.947
Soft GDMA	0.843	0.943	1.028	0.20	0.06	0.978	0.878	0.964
Fair soft GDMA	0.840	0.937	1.022	0.19	0.03	0.977	0.875	0.960
Areas	125						88	37

^d Highest continent-level median of the per-area ratios. ^e Areas with less than two years of evaluation data (the 31 Chinese provinces and Algeria).

shared rule, with unit-level shrinkage captures most of the heterogeneity with far less noise, and the segmentation can be discussed with operators. The evidence for grouping itself is modest: it is statistically indistinguishable from simpler clustering rules. Third, more flexible is not better: a neural mixture-of-experts gate with thousands of parameters was seven percentage points more expensive worldwide than soft GDMA and made 42% of the grid areas worse off than their own best candidate. Fourth, fairness in the sense of the criterion was nearly free, whereas maximin storage siting cost 7–9% of the programme’s value in price of fairness. Fifth, forecasting and investment are linked: in a stylised model, the reference rule needs 34 GW of optimally sited batteries to match the cost that soft GDMA achieves without any, and a programme sited on today’s error profile loses 2–4% of its value; this is an interaction to study, not a siting recommendation. Because GDMA only re-weights existing forecasts, it can be adopted without replacing forecasting systems.

Who would adopt it. The grouping needs someone who sees many units at once: a regional transmission organisation computing rules for its member balancing authorities, European regional coordination centres, a multi-region market operator such as AEMO, or forecasting-service providers and regulators that work with many utilities. The data required are each unit’s candidate commitments and realised demand, which such coordinators typically hold; units that cannot share them could exchange only cost gradients, as in federated learning. The rules are auditable, as each commitment is a convex combination of commitments the unit already understands. GDMA feeds the day-ahead commitment into existing processes and does not replace reserve procurement or unit commitment.

Limitations and extensions. Coverage of Africa and parts of Asia is thin, because public hourly demand data for South Africa, India, Japan and South Korea could not be retrieved and the Chinese provinces are observed in a single year; such data-poor systems benefit much less. Income is measured for countries and U.S. states, not households. Outside

Table 15: Benefits by income ($W = 28$): commitment cost relative to the reference forecaster by World Bank income group and, within the United States, for the balancing authorities in the lowest and highest third of state per-capita income; Spearman rank correlations between an area’s cost ratio and income. Lower is better: values below 1 mean a lower cost than the benchmark.

Method	All 125 areas			U.S. balancing authorities		
	High income	Upper-middle income	Rank corr. with GDP p.c. ^g	Lowest income third	Highest income third	Rank corr. with state income ^g
Reference forecaster ^a	1.000	1.000	–	1.000	1.000	–
Equal weights	1.235	1.014	0.530	1.636	1.141	-0.106
Best single (per area)	0.917	0.992	-0.126	0.964	0.692	0.013
Pooled DF weights	0.901	0.918	0.185	0.959	0.729	-0.033
Per-area DF weights	0.888	0.969	-0.164	0.941	0.680	0.015
Shrinkage to pooled	0.883	0.947	-0.093	0.936	0.680	0.009
Two-step k -means	0.880	0.928	0.108	0.938	0.696	0.063
Quantile-regression comb.	1.011	1.074	-0.204	1.128	0.771	-0.107
Forecast-then-commit, per area	0.983	1.043	-0.010	1.101	0.768	-0.122
Forecast-then-commit, grouped	0.962	1.011	0.046	1.089	0.781	-0.122
Neural gate (deep MoE)	0.953	0.989	0.086	1.128	0.793	-0.127
GDMA	0.879	0.938	0.025	0.931	0.689	0.045
Soft GDMA	0.878	0.951	-0.120	0.927	0.678	0.028
Fair soft GDMA	0.875	0.947	-0.114	0.927	0.678	0.039
Areas	83	42		15	16	

^g A negative correlation means that richer areas have lower cost ratios, i.e. gain more.

the United States the reference is a model forecast, because official day-ahead forecasts of European transmission system operators and of AEMO could not be retrieved, and in the United States the reference is a stylised policy, the official forecast plus a rolling quantile margin. Shortfall costs rest on a proxy based on the variability of interchange, which is not transfer capability, rather than on market prices or the value of lost load, and the single-period newsvendor ignores ramping, network constraints and multi-day commitments, and demand served in a load-shedding event understates true demand. The weather candidates use archived GFS and GEFS forecasts at 0.5–1 degree resolution, from late 2020 only and only where the 00 UTC run is available before gate closure; they roughly halve the gap to the operators’ forecasts during Winter Storm Elliott, but the combinations remain about 10% more expensive than the operators over those days, even though one candidate designed with that storm in view is 7% cheaper on its own: weights learned from previous weeks cannot anticipate a storm. The storage stage takes the commitment rule as given and values only commitment errors, so joint optimisation and other value streams could change where storage should go. The theory treats the number of segments as given and relies on a high-level concentration condition; the hold-out choice of G and κ and strong cross-sectional dependence through common weather shocks are left for future work.

Conclusion. Grouped decision-focused model averaging combines optimal model averaging, decision-focused learning and latent group structures in one estimator with guarantees for the idealised version, and links it to a stylised siting problem. Using only information available at the day-ahead gate closure, it turns forecasts operators already have into commitment rules that are 12% cheaper worldwide, and 13–17% cheaper in the United States, than committing on a reference forecaster; the gain comes mostly from deciding for the decision and sharing structure across units, with grouping a smaller refinement. The limits matter too: no rule beats the operators during an abrupt storm, entering units

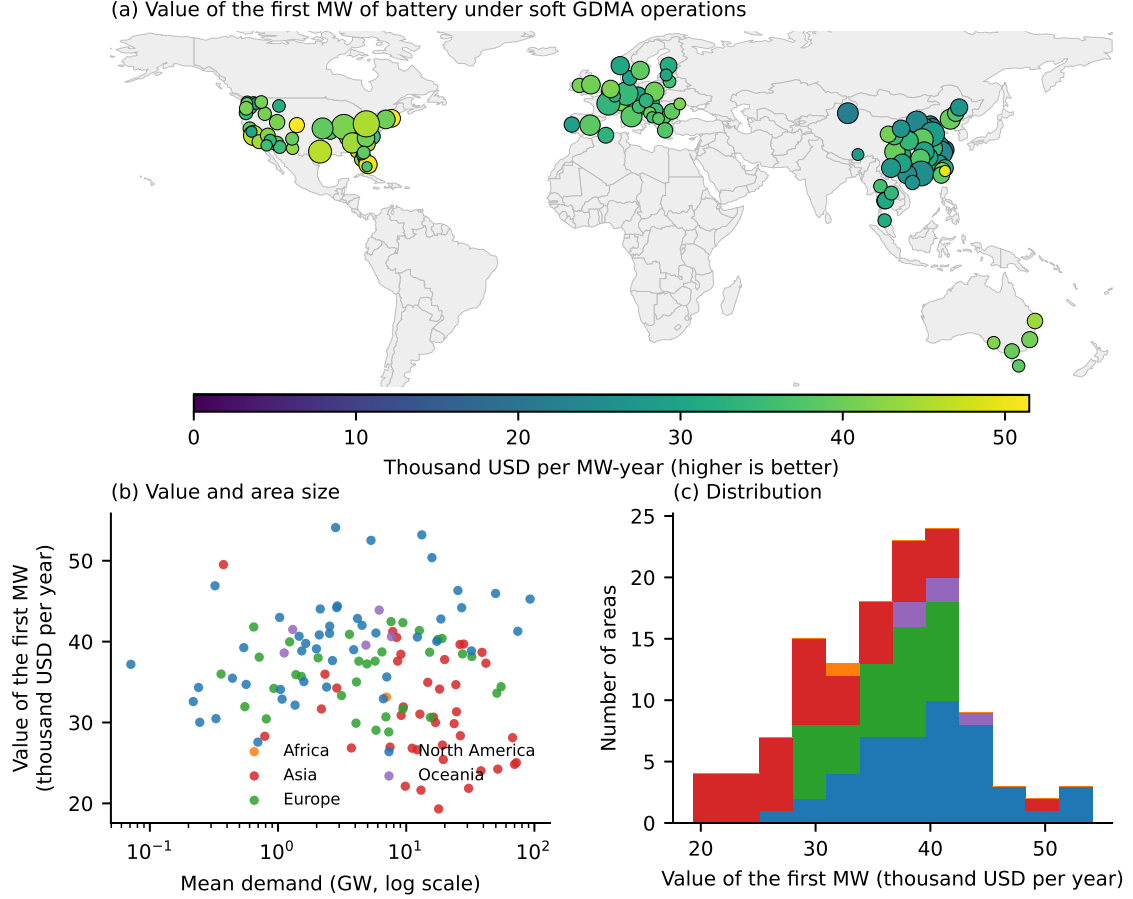


Figure 7: Where batteries are worth most when operations follow soft GDMA: value of the first MW of battery power in each area (higher is better), against the area's size and its distribution.

are poorly served by every rule, and the value of better commitment for storage planning depends on value streams we do not model.

Declaration of competing interest

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Quang-Vinh Dang: Conceptualization, Methodology, Software, Formal analysis, Data curation, Investigation, Visualization, Writing – original draft, Writing – review & editing.

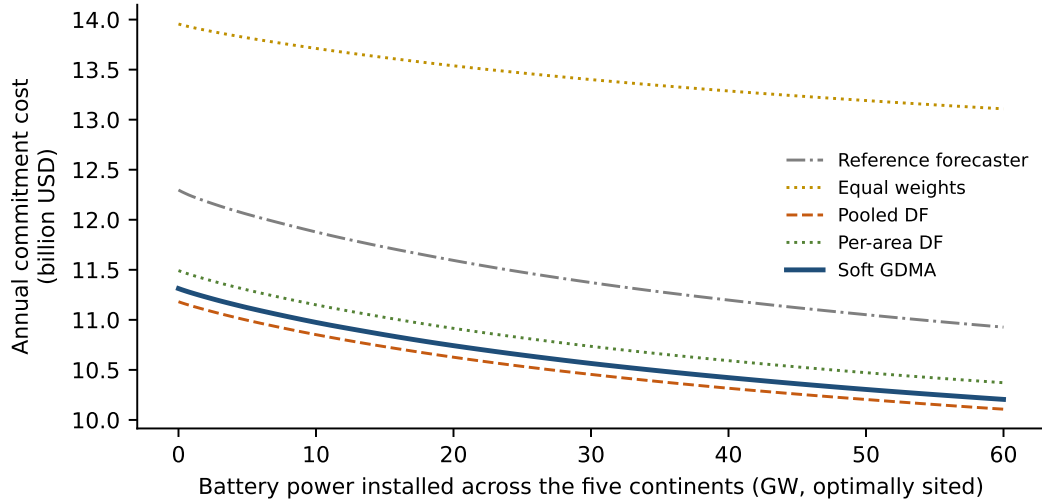


Figure 8: Annual system-wide commitment cost against battery power installed and optimally sited across the 125 grid areas, for five commitment rules. Costs valued at \$10 per MWh of idle committed capacity. Lower is better.

Data availability

All data used in this study are publicly available from the sources cited in Sections 6 and 7. The code that downloads the data, builds the panels and reproduces every table and figure is available at https://github.com/vinhqdang/economics_data_science.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used generative AI tools to support the writing of code and to polish the language of the manuscript. The ideas, the data and the results are the author’s own. The author reviewed and edited all output and takes full responsibility for the content of the publication.

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Table 16: Storage siting on five continents. Cost of each rule without storage, battery power it needs (optimally sited) to reach the cost of soft GDMA without storage, and optimal siting of a 10 GW programme. Cost: lower is better; battery power: lower means the rule needs less storage to reach the same cost.

Commitment rule	Cost without storage (\$bn/yr)	GW to match soft GDMA ^c	Siting of a 10 GW budget (GW)				
			N. Am.	Europe	Oceania	Asia	Africa
Reference forecaster	12.30	34	5.1	1.6	0.4	2.9	0.0
Equal weights	13.96	> 60	1.5	1.8	1.2	5.5	0.1
Pooled DF	11.18	0	4.1	2.6	0.7	2.5	0.0
Per-area DF	11.49	5	4.3	2.6	0.7	2.5	0.0
Forecast-then-commit	11.92	17	5.0	2.4	0.4	2.1	0.0
GDMA	11.21	0	4.3	2.6	0.7	2.4	0.0
Soft GDMA	11.31	–	4.3	2.6	0.7	2.5	0.0

^c Smallest optimally sited budget, on a grid of 0.25 GW steps up to 5 GW and 1 GW steps up to 60 GW, at which the rule’s cost falls to or below that of soft GDMA without storage.

Table 17: Annual value (million USD) of storage programmes of different sizes when operations follow soft GDMA, by the rule whose error profile was used to site the batteries. Higher is better.

Siting based on	1 GW	5 GW	10 GW	20 GW
Reference forecaster	43	184	327	560
Per-area DF	45	191	338	570
Forecast-then-commit	44	187	333	566
Soft GDMA	45	191	338	571

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Table 18: Out-of-sample siting under soft GDMA operations (million USD per year): batteries are sited on the value curves of the first half of the evaluation period and valued on the second half, and compared with siting on the second half itself (infeasible oracle). Higher is better.

Budget (GW)	Sited on soft GDMA (first half)	Sited on the reference (first half)	Sited on the second half itself (oracle)
1	44	42	47
5	148	145	194
10	247	232	336
20	391	360	539

Table 19: Efficient and fair siting of storage programmes under soft GDMA operations: annual value, the relative benefit of the worst-served continent, and the price of fairness (PoF, the share of the efficient programme’s value given up). Value and relative benefit: higher is better; price of fairness: lower is better.

Budget (GW)	Efficient		Proportional floors ($\lambda = 1$)			Maximin		
	Value (\$m/yr)	Min. rel. benefit ^f	Value (\$m/yr)	Min. rel. benefit	PoF	Value (\$m/yr)	Min. rel. benefit	PoF
1	45	0.0%	42	0.2%	7.3%	41	0.3%	9.0%
5	191	0.6%	181	0.9%	5.0%	178	1.5%	6.9%
10	338	1.4%	325	1.7%	3.9%	315	2.8%	7.0%
20	571	3.1%	554	3.3%	3.0%	532	4.7%	6.7%

^f Lowest, across continents, of the value captured divided by the continent’s annual commitment cost without storage.

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Appendix A. Proofs

Throughout, $\|\cdot\|_1$ is the ℓ_1 norm and constants c, C may change from line to line.

Covering the simplex. For $\epsilon \in (0, 1]$ there is a set $\mathcal{N}_\epsilon \subset \Delta_M$ with $|\mathcal{N}_\epsilon| \leq (3/\epsilon)^M$ such that every $\mathbf{w} \in \Delta_M$ has a point $\mathbf{w}' \in \mathcal{N}_\epsilon$ with $\|\mathbf{w} - \mathbf{w}'\|_1 \leq \epsilon$ (volumetric argument for the ℓ_1 ball, e.g. [Vershynin, 2018](#), Ch. 4). Hence $\mathcal{N}_\epsilon^G$ is an ϵ -net of Δ_M^G in the norm $\max_g \|\mathbf{w}_g - \mathbf{w}'_g\|_1$, of cardinality at most $(3/\epsilon)^{GM}$.

Lipschitz property. For all $(\boldsymbol{\gamma}, \mathbf{W})$ and $(\boldsymbol{\gamma}, \mathbf{W}')$ with $\max_g \|\mathbf{w}_g - \mathbf{w}'_g\|_1 \leq \epsilon$, Assumption 1 gives $|\hat{R}(\boldsymbol{\gamma}, \mathbf{W}) - \hat{R}(\boldsymbol{\gamma}, \mathbf{W}')| \leq \bar{a}B\epsilon$ and $|R(\boldsymbol{\gamma}, \mathbf{W}) - R(\boldsymbol{\gamma}, \mathbf{W}')| \leq \bar{a}B\epsilon$.

Proof of Theorem 1

Fix $\epsilon = 1/(NT_e)$. The set $\mathcal{F} = \{1, \dots, G\}^N \times \mathcal{N}_\epsilon^G$ has cardinality at most $G^N(3NT_e)^{GM}$. By Assumption 2 and the union bound,

$$\Pr \left(\max_{(\boldsymbol{\gamma}, \mathbf{W}) \in \mathcal{F}} |\hat{R}(\boldsymbol{\gamma}, \mathbf{W}) - R(\boldsymbol{\gamma}, \mathbf{W})| > x \right) \leq 2G^N(3NT_e)^{GM} \exp(-cNT_e x^2/L^2).$$

Setting the right-hand side equal to δ gives, with probability at least $1 - \delta$, $\max_{\mathcal{F}} |\hat{R} - R| \leq x_\delta := L\{(GM \log(3NT_e) + N \log G + \log(2/\delta))/(cNT_e)\}^{1/2}$. For an arbitrary $(\boldsymbol{\gamma}, \mathbf{W}) \in \mathcal{P}_G$ let $\mathbf{W}'$ be its nearest net point; by the Lipschitz property,

$$\sup_{\mathcal{P}_G} |\hat{R} - R| \leq x_\delta + 2\bar{a}B\epsilon.$$

Let $(\boldsymbol{\gamma}^*, \mathbf{W}^*)$ satisfy $R(\boldsymbol{\gamma}^*, \mathbf{W}^*) \leq R_G^* + \eta$ for arbitrary $\eta > 0$. Since $(\hat{\boldsymbol{\gamma}}, \widehat{\mathbf{W}})$ minimises $\hat{R}$,

$$R(\hat{\boldsymbol{\gamma}}, \widehat{\mathbf{W}}) \leq \hat{R}(\hat{\boldsymbol{\gamma}}, \widehat{\mathbf{W}}) + x_\delta + 2\bar{a}B\epsilon \leq \hat{R}(\boldsymbol{\gamma}^*, \mathbf{W}^*) + x_\delta + 2\bar{a}B\epsilon \leq R_G^* + \eta + 2x_\delta + 4\bar{a}B\epsilon.$$

Letting $\eta \downarrow 0$ and substituting $\epsilon = 1/(NT_e)$ gives (5). $\square$

Proof of Corollary 1

Theorem 1 with $\delta = \delta_{N,T} \rightarrow 0$ slowly (for instance $\log(1/\delta) = \log(NT_e)$) shows $R(\hat{\boldsymbol{\gamma}}, \widehat{\mathbf{W}}) - R_G^* = O_p(r_{N,T})$ with $r_{N,T} = \{GM \log(NT_e)/(NT_e) + \log G/T_e + \log(NT_e)/(NT_e)\}^{1/2} \rightarrow 0$. Dividing by $R_G^* \geq \kappa$ yields $R(\hat{\boldsymbol{\gamma}}, \widehat{\mathbf{W}})/R_G^* - 1 = O_p(r_{N,T}/\kappa) \rightarrow_p 0$. Because $R(\hat{\boldsymbol{\gamma}}, \widehat{\mathbf{W}}) \geq R_N^*$ always and $R_G^* \geq R_N^*$, the second claim follows once we show $R_G^* = R_N^*$ under the stated structure. Let $\mathbf{w}_g^0$ be a common minimiser of R_i for all i with $\gamma_i^0 = g$ (padding with arbitrary vectors if $G_0 < G$); then $R(\boldsymbol{\gamma}^0, \mathbf{W}^0) = N^{-1} \sum_i \min_{\mathbf{w}} R_i(\mathbf{w}) = R_N^*$, so $R_G^* \leq R_N^*$, and the reverse inequality holds because $\mathcal{P}_G$ is a subset of the unit-specific class. $\square$

Proof of Theorem 2

Step 1 (average consistency). Under Assumption 3(a), $R_G^* = R(\boldsymbol{\gamma}^0, \mathbf{W}^0) = R_N^*$ and

$$\frac{\lambda}{N} \sum_{i=1}^N \left\| \widehat{\mathbf{w}}_{\widehat{\gamma}_i} - \mathbf{w}_{\gamma_i^0}^0 \right\|_2^2 \leq R(\widehat{\boldsymbol{\gamma}}, \widehat{\mathbf{W}}) - R_G^* = o_p(1)$$

by Theorem 1, since $\log N/T_e \rightarrow 0$ implies $\log G/T_e \rightarrow 0$ and $GM \log(NT_e)/(NT_e) \rightarrow 0$ for fixed G and M .

Step 2 (consistency of the segment weights). Fix $\eta \in (0, d/4)$. By Step 1 and Markov's inequality, the fraction of units with $\|\widehat{\mathbf{w}}_{\widehat{\gamma}_i} - \mathbf{w}_{\gamma_i^0}^0\|_2 > \eta$ is $o_p(1)$. Since every true segment g contains at least πN units, with probability approaching one each g contains a unit i with $\|\widehat{\mathbf{w}}_{\widehat{\gamma}_i} - \mathbf{w}_g^0\|_2 \leq \eta$; call $h(g) = \widehat{\gamma}_i$. If $h(g) = h(g')$ for $g \neq g'$, the triangle inequality gives $\|\mathbf{w}_g^0 - \mathbf{w}_{g'}^0\|_2 \leq 2\eta < d$, contradicting Assumption 3(b). Thus h is injective, hence a bijection of $\{1, \dots, G\}$, and after relabelling $\max_g \|\widehat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \leq \eta$ with probability approaching one. Since η is arbitrary, (i) follows.

Step 3 (memberships). On the event $E_\eta = \{\max_g \|\widehat{\mathbf{w}}_g - \mathbf{w}_g^0\|_2 \leq \eta\}$, unit i with $\gamma_i^0 = g$ can be assigned to $h \neq g$ only if $\widehat{R}_i(\widehat{\mathbf{w}}_h) \leq \widehat{R}_i(\widehat{\mathbf{w}}_g)$; every global minimiser assigns each unit to a segment minimising its own empirical cost, since otherwise reassigning the unit would lower (2). By Assumption 3 and the Lipschitz property (with $\|\cdot\|_1 \leq \sqrt{M}\|\cdot\|_2$),

$$R_i(\widehat{\mathbf{w}}_h) - R_i(\widehat{\mathbf{w}}_g) \geq \lambda(d - \eta)^2 - \bar{a}B\sqrt{M}\eta \geq \lambda d^2/4 =: \Delta$$

for η small enough. Hence misclassification of unit i on E_η requires $\sup_{\mathbf{w} \in \Delta_M} |\widehat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| \geq \Delta/2$. Covering Δ_M by $\mathcal{N}_\epsilon$ with $\epsilon = \Delta/(8\bar{a}B)$ and applying the unit-level part of Assumption 2,

$$\Pr \left(\sup_{\mathbf{w}} |\widehat{R}_i(\mathbf{w}) - R_i(\mathbf{w})| \geq \Delta/2 \right) \leq 2 \left(\frac{24\bar{a}B}{\Delta} \right)^M \exp \left(- \frac{cT_e \Delta^2}{16L^2} \right) =: C_M e^{-c'T_e}.$$

Because the supremum is over all $\mathbf{w}$, the bound does not require independence between $\widehat{\mathbf{W}}$ and unit i 's data. A union bound over units gives $\Pr(\widehat{\boldsymbol{\gamma}} \neq \boldsymbol{\gamma}^0) \leq \Pr(E_\eta^c) + NC_M e^{-c'T_e}$, which tends to zero because $\log N/T_e \rightarrow 0$. This is (ii).

Step 4 (oracle equivalence). On the event $\{\widehat{\boldsymbol{\gamma}} = \boldsymbol{\gamma}^0\}$, $\widehat{\mathbf{W}}$ minimises $\widehat{R}(\boldsymbol{\gamma}^0, \cdot)$, so $\widehat{\mathbf{W}} = \widetilde{\mathbf{W}}$ (if the minimiser is not unique, the two estimators select from the same set). By (ii) this event has probability approaching one. $\square$

Appendix B. Out-of-sample risk

The theory concerns the population risk R evaluated at the estimated policy. In a rolling application the policy estimated on periods $1, \dots, T$ is applied in periods $T + 1, \dots, T + k$. If $\{\mathbf{z}_{it}\}$ is β -mixing with coefficients $\beta(\cdot)$, a coupling argument (Yu, 1994) shows that the expected cost of the estimated policy in period $T + s$ differs from $\mathbb{E} R(\widehat{\boldsymbol{\gamma}}, \widehat{\mathbf{W}})$ by at most $L\beta(s)$. The bounds of Section 4 therefore also bound the out-of-sample cost up to a term that vanishes with the gap between estimation and application.

Appendix C. Balancing authorities

Appendix D. Additional results for the global panel

Table C.20: The 46 balancing authorities, ordered by mean hourly demand over 2019–2026. Flex.: interchange flexibility in 2019 (range between the 5th and 95th percentiles of hourly total interchange, divided by mean demand). τ_i : calibrated critical ratio.

Code	Balancing authority	Mean MW	Flex.	τ_i
PJM	PJM Interconnection, LLC	92,507	0.06	0.947
MISO	Midcontinent Independent System Operator, Inc.	74,189	0.07	0.943
ERCO	Electric Reliability Council of Texas, Inc.	49,750	0.03	0.950
SWPP	Southwest Power Pool	32,259	0.08	0.940
SOCO	Southern Company Services, Inc. - Trans	27,075	0.12	0.927
CISO	California Independent System Operator	25,319	0.29	0.900
TVA	Tennessee Valley Authority	18,637	0.09	0.937
NYIS	New York Independent System Operator	17,306	0.18	0.910
FPL	Florida Power & Light Co.	15,873	0.10	0.933
ISNE	ISO New England	13,228	0.16	0.920
DUK	Duke Energy Carolinas	12,212	0.23	0.907
CPLE	Duke Energy Progress East	7,045	0.37	0.880
BPAT	Bonneville Power Administration	6,631	0.84	0.827
PACE	PacifiCorp East	5,803	0.42	0.877
PSCO	Public Service Company of Colorado	5,312	0.11	0.930
NEVP	Nevada Power Company	4,523	0.33	0.883
LGEE	Louisville Gas and Electric Company and Kentucky Utilities Company	4,166	0.17	0.913
AZPS	Arizona Public Service Company	3,893	0.67	0.843
SC	South Carolina Public Service Authority	2,901	0.32	0.890
LDWP	Los Angeles Department of Water and Power	2,867	0.30	0.897
SCEG	Dominion Energy South Carolina, Inc.	2,814	0.15	0.923
AECI	Associated Electric Cooperative, Inc.	2,658	0.52	0.867
TEC	Tampa Electric Company	2,530	0.26	0.903
PGE	Portland General Electric Company	2,523	0.53	0.863
PACW	PacifiCorp West	2,392	0.55	0.860
FMPP	Florida Municipal Power Pool	2,135	0.17	0.917
IPCO	Idaho Power Company	2,097	0.79	0.833
BANC	Balancing Authority of Northern California	1,993	0.59	0.853
PNM	Public Service Company of New Mexico	1,645	0.64	0.847
TEPC	Tucson Electric Power	1,591	0.71	0.837
JEA	JEA	1,532	0.32	0.893
AVA	Avista Corporation	1,460	0.70	0.840
NWMT	NorthWestern Corporation	1,358	1.07	0.810
SCL	Seattle City Light	1,076	0.92	0.817
WALC	Western Area Power Administration - Desert Southwest Region	1,045	0.95	0.813
EPE	El Paso Electric Company	1,029	0.46	0.873
GCPD	Public Utility District No. 2 of Grant County, Washington	696	1.58	0.807
CPLW	Duke Energy Progress West	564	0.61	0.850
TPWR	City of Tacoma, Department of Public Utilities, Light Division	540	0.56	0.857
IID	Imperial Irrigation District	441	0.90	0.820
TIDC	Turlock Irrigation District	328	0.81	0.830
TAL	City of Tallahassee	322	0.32	0.887
DOPD	PUD No. 1 of Douglas County	245	2.21	0.803
GVL	Gainesville Regional Utilities	240	0.50	0.870
CHPD	Public Utility District No. 1 of Chelan County	218	4.71	0.800
HST	City of Homestead	71	0.89	0.823

Table D.21: Global panel with the main candidate set (eleven candidates including the weather models): system cost relative to the reference forecaster for estimation windows of 14 and 28 days (world). Lower is better: values below 1 mean a lower cost than the benchmark.

Method	$W = 14$	$W = 28$
Reference forecaster ^a	1.000	1.000
Equal weights	1.217	1.217
Best single (per area)	0.943	0.923
Pooled DF weights	0.906	0.902
Per-area DF weights	0.918	0.895
Shrinkage to pooled	0.905	0.888
Two-step k -means	0.899	0.884
Quantile-regression comb.	1.174	1.016
Forecast-then-commit, per area	1.030	0.988
Forecast-then-commit, grouped	0.990	0.966
Neural gate (deep MoE)	–	0.956
GDMA	0.900	0.884
Soft GDMA	0.902	0.884
Fair soft GDMA	0.897	0.881

Table D.22: Global panel with the main candidate set and with a common critical ratio $\tau = 0.9$ for every grid area ($W = 28$): system cost relative to the reference forecaster. The neural gate was not run for this specification. Lower is better: values below 1 mean a lower cost than the benchmark.

Method	World	North America	Europe	Oceania	Asia	Africa
Reference forecaster ^a	1.000	1.000	1.000	1.000	1.000	1.000
Equal weights	1.193	1.401	1.059	0.996	1.035	0.976
Best single (per area)	0.917	0.854	0.942	1.008	1.023	0.915
Pooled DF weights	0.899	0.863	0.914	0.983	0.935	0.926
Per-area DF weights	0.891	0.832	0.912	0.989	0.990	0.932
Shrinkage to pooled	0.884	0.830	0.905	0.984	0.971	0.925
Two-step k -means	0.879	0.833	0.896	0.975	0.946	0.895
Quantile-regression comb.	0.987	0.945	0.987	0.983	1.179	1.064
Forecast-then-commit, per area	0.958	0.934	0.932	0.989	1.184	0.946
Forecast-then-commit, grouped	0.938	0.932	0.898	0.981	1.145	0.926
GDMA	0.878	0.827	0.899	0.973	0.954	0.891
Soft GDMA	0.880	0.824	0.901	0.976	0.972	0.908
Fair soft GDMA	0.876	0.822	0.896	0.976	0.969	0.908